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**EVEREST – An
Extreme-Value/Evidential-Retentive
Transformer for Operational Solar-Flare
Forecasting**

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Design Engineering Master's Project

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part of the MEng degree in Design Engineering.

Declaration of Originality

I hereby declare that, unless explicitly stated otherwise, the work presented in this thesis is the result of my own efforts. To the best of my knowledge, this work is original, and any contributions or ideas developed in collaboration with others have been duly acknowledged and referenced. In the course of this project, Large Language Models (LLMs) were employed to support various aspects of the research. These tools—including Grammarly and ChatGPT—were used to enhance clarity and fluency in written expression, assist in project ideation, and, where relevant, form part of the technical workflow.

—Antanas Žilinskas, June 5, 2025

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Abstract

Solar flares threaten critical infrastructure and human activities, highlighting the necessity for reliable long-range forecasting systems. This thesis presents EVEREST (Evidential and EVT-enhanced Rare-Event Sequence Transformer), a novel deep learning architecture that enhances solar flare prediction, focusing on high-impact, rare events up to 72-hour forecasts. EVEREST features a learnable attention bottleneck for refined temporal feature extraction, coupled with a comprehensive uncertainty quantification framework. This includes an evidential (Normal-Inverse-Gamma) head for calibrated probabilistic outputs and an Extreme Value Theory (EVT) head for tail-risk modelling. A multi-objective composite loss function, incorporating focal loss, addresses severe class imbalance and effectively guides the learning process.

Trained and validated on over a decade of SHARP vector magnetogram data from the Solar Dynamics Observatory and GOES flare catalogues, EVEREST demonstrates improved predictive performance over the previous state-of-the-art model, setting new benchmarks across flare classes (C, M, M5) and forecast windows. Notably, it achieves a True Skill Statistic (TSS) of 0.966 for M5-class flares at a 72-hour lead time, coupled with effective probabilistic calibration (e.g., Expected Calibration Error of 0.016). Interpretability analyses reveal that EVEREST’s attention mechanisms and learned features match known heliophysical precursor signatures, including magnetic flux emergence and polarity inversion line shear. The model’s practical utility is validated through its performance in a prospective case study of the September 2017 X9.3 flare, offering operationally relevant lead times. Further validation on the SKAB anomaly detection benchmark indicates that EVEREST’s core principles may apply broadly to other rare-event time-series forecasting domains.

This research contributes a principled and high-performing deep learning framework for operational solar flare forecasting. EVEREST enhances predictive accuracy and reliability, paving the way for more trustworthy AI in space weather science. Future efforts will prioritise multimodal data integration, federated learning strategies, and model compression for edge deployment, thereby enhancing capabilities and operational readiness.

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Publications

The following publications were derived from work undertaken during this project:

- A. Žilinskas, R. N. Shorten, and J. Marecek, EVEREST: A Transformer for Probabilistic Rare-Event Anomaly Detection with Evidential and Tail-Aware Uncertainty, Submitted to *Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- A. Žilinskas, Solar Flare Prediction Dataset: SHARP Magnetic Field Parameters for Machine Learning, *Zenodo*, 2025, doi: [10.5281/zenodo.15570399](https://doi.org/10.5281/zenodo.15570399).
- A. Žilinskas, EVEREST: Source Code, *GitHub Repository*, 2025, <https://github.com/AntanasZilinskas/masters-project>.

Chapter 1

Introduction

Solar flares release up to 10^{32} erg of magnetic energy in the low corona (Benz, 2016). The resulting extreme-ultraviolet (EUV) and X-ray photons rapidly disrupt Earth’s ionosphere, affecting HF radio, GNSS, and polar aviation. Associated coronal mass ejections (CMEs) can induce damaging currents in power grids and satellites; for instance, the 1989 Hydro-Québec blackout cost $\sim$ USD 2 billion (Kappenman, 2010). Although X-class flares are rare ($<0.03\%$ of observations across six solar cycles), their substantial socio-economic impact necessitates reliable forecasts for agencies such as the NOAA Space Weather Prediction Center (SWPC).

Operational flare categories. The Geostationary Operational Environmental Satellite (GOES) programme labels flares by their 1–8 Å peak soft-X-ray flux. Thresholds and indicative impacts are summarised in Table 1.1. This study concentrates on events of class C and above, which pose the greatest operational risk.

Table 1.1: GOES soft-X-ray flare classes and representative impacts.

Class	Peak flux (W m^{-2})	Indicative impact
A, B	$< 10^{-7}$	none
C	10^{-6} – 10^{-5}	minor HF fade (high latitudes)
M	10^{-5} – 10^{-4}	polar radio blackout; proton storm onset
X	$\geq 10^{-4}$	global HF blackout; energetic particle showers

Limits of current machine-learning systems. Support-vector and random-forest classifiers built on SHARP physical parameters plateau at a 24-h $\geq$ M true-skill statistic (TSS) of about 0.55 (Bobra and Couvidat, 2015), a ceiling confirmed in solar-cycle-25 benchmarks (Kazachenko et al., 2022; Murray et al., 2023). Convolutional–recurrent hybrids improve recall yet remain limited to 24 h owing to quadratic attention or gated

memory (Huang et al., 2018; Sun, Bobra, Gombosi and et al., 2022). Long-context Transformers such as MEGA (Zhou et al., 2021) widen receptive fields, but SolarFlareNet (Abduallah et al., 2024) and other CNN+LSTM methods (Sun, Bobra, Gombosi and et al., 2022) truncate inputs to fit GPU memory and therefore peak at a 48 h prediction horizon.

From SolarKnowledge to EVEREST

Transformer architectures, such as *Informer* (Zhou et al., 2020), *PatchTST* (Nie et al., 2024), and MEGA (Ma et al., 2023), are prominent in time-series forecasting. Our initial Transformer baseline, *SolarKnowledge* (six standard self-attention blocks (Vaswani et al., 2017) with focal loss and Monte-Carlo dropout), achieved a 24-hour $\geq$ M True Skill Statistic (TSS) of 0.87 but degraded sharply beyond 48 hours. These shortcomings motivated the **Evidential and EVT-enhanced Rare-Event Sequence Transformer** (*EVEREST*). EVEREST incorporates: (i) a learnable attention bottleneck for extended temporal windows (up to 120 hours or 432 tokens); (ii) an integrated uncertainty quantification framework with dual heads for Normal-Inverse-Gamma evidential learning (calibrated probabilities) and Generalised Pareto Distribution-based Extreme Value Theory modelling (explicit tail-risk assessment); and (iii) a precursor-probability auxiliary head to enhance early warning signals. On the SHARP-GOES benchmark, for $\geq$ M5.0 class flares, EVEREST achieves TSS scores of 0.907 (24h), 0.936 (48h), and 0.966 (72h), demonstrating state-of-the-art performance.

The main contributions of this research include:

- (a) EVEREST, a novel architecture achieving state-of-the-art long-range solar flare forecasting (e.g., TSS of 0.966 for M5-class flares at 72h).
- (b) Demonstrated probabilistic calibration (e.g., Expected Calibration Error of 0.016 for M5-72h flares), yielding trustworthy uncertainty estimates for operational use.
- (c) A documented, open-source EVEREST implementation (Žilinskas, 2025) with auditable MLOps, energy metrics, and reproducible artifacts, promoting transparency and further research.

Report structure. Subsequent chapters describe the dataset (Chapter 3), model architecture (Chapter 4), evaluation (Chapter 5), operational implications (Chapter 6), and conclusions (Chapter 9).

Chapter 2

Literature Review and Contextual Investigation

Solar flares, originating in magnetically complex active regions (ARs), release vast energy as magnetic reconnection converts stored magnetic energy into thermal, kinetic, and radiative forms ([Priest and Forbes, 2002](#); [Shibata and Magara, 2011](#)).

Impending instability in flare-productive ARs is often indicated by strong magnetic shear, new flux emergence, and complex (β – γ – δ) sunspot configurations ([Toriumi and Wang, 2019](#)), reflecting nonpotential energy accumulation released via reconnection ([Janvier, 2015](#)). Reconnection, forming at current sheets or topological discontinuities (null points, quasi-separatrix layers), triggers rapid restructuring that accelerates particles and produces broad-spectrum emissions (e.g., soft X-rays to radio waves) ([Benz, 2016](#)).

Coronal reconnection exhibits measurable photospheric precursors. The Helioseismic and Magnetic Imager (HMI) aboard NASA’s Solar Dynamics Observatory captures vector magnetograms quantifying field evolution ([Scherrer, 2012](#)). From these, the Space-weather HMI Active Region Patch (SHARP) product extracts over 40 parameters (e.g., current helicity, total unsigned flux, vertical current density) that proxy magnetic complexity and flare potential ([Bobra, 2014](#)). While Parker Solar Probe measurements constrain inner heliosphere reconnection rates, they do not replace photospheric precursors for Earth-directed forecasting ([Fox et al., 2024](#)). SHARP data thus remain foundational for modern machine-learning (ML) flare predictors ([Leka, 2018](#)).

2.1 Rare-Event Forecasting in Machine Learning: Sequence Models and Imbalance Challenges

From a statistical-learning viewpoint, flare prediction is a prototypical rare-event classification task. M- and X-class flares represent less than 0.5% of magnetogram dataset windows (Barnes, 2016), resulting in a significant imbalance that skews naïve classifiers towards "all-clear" outputs (Nishizuka, 2017).

Initial ML approaches employed snapshot classifiers (e.g., SVMs) on SHARP features, achieving moderate TSS (0.5–0.6 for 24-h >M forecasts) but neglecting AR temporal evolution (Bobra and Couvidat, 2015; Bloomfield, 2012; Muranushi, 2015). Subsequent sequence-aware models like LSTMs (Chen, 2019) and TCNs (Nishizuka et al., 2018) offered improvements but faced inherent limitations such as vanishing gradients (LSTMs) or the need for careful dilation scheduling (TCNs) for multi-day horizons.

Transformers for time series. Self-attention (Vaswani et al., 2017) enables Transformers to handle distant timesteps, outperforming recurrent architectures. While models like TimesNet (Chen et al., 2023) and PatchTST++ (Nie et al., 2024) excel on general benchmarks, vanilla self-attention’s $\mathcal{O}(T^2)$ complexity restricts input windows in flare studies to ≤ 72 h (Sun, Bobra, Gombosi and Munoz-Jaramillo, 2022). Efficient methods like FlashAttention-2 (Dao et al., 2023) require custom CUDA kernels, limiting solar-physics adoption. For rarer $\geq$ M5.0 flares (24 h horizon), reported TSS ranges from 0.818 (SolarFlareNet (Abdullah et al., 2023)) to 0.881 (GRU baseline Liu et al. (2019)); Sun, Bobra, Gombosi and et al. (2022) excludes this tier. Meanwhile, single-head soft-attention pooling, derived from multiple-instance learning and Set Transformers, enables $\mathcal{O}(T)$ sequence summarization without specialised kernels (Ilse et al., 2018; Lee et al., 2019), thus reducing computational load for extended sequences.

Standard binary cross-entropy is suboptimal for rare-event forecasting due to class scarcity. Focal loss (Lin et al., 2017b; Abdullhah et al., 2023) mitigates this by down-weighting well-classified examples, improving rare-flare recall. However, operational utility also demands reliable, statistically calibrated probabilities, as now mandated by NOAA’s SWPC to ensure forecast integrity (i.e., predicted probabilities matching observed frequencies) (NOAA SWPC, 2023). Evidential learning (Sensoy et al., 2018) quantifies aleatoric and epistemic uncertainty, often by predicting a distribution (e.g., Normal-Inverse-Gamma) over logits, offering richer uncertainty insights. These outputs must still provide or be convertible to calibrated probabilities, crucial for robust decision-making in space weather forecasting (Pinson and Tastu, 2021; NOAA Space Weather Prediction Center, 2022).

Tail awareness. Flare energies obey a power-law distribution (Aschwanden, 2011). Standard losses optimise average error and under-represent catastrophic extremes. Extreme-Value Theory models distributional tails with the Generalised Pareto Distribution (GPD) (Coles, 2001); recent neural implementations add an EVT penalty that emphasises exceedances (Gissibl et al., 2022). Hybrid EVT-attention models have emerged in 2024 for hurricane wind-burst prediction, highlighting EVT’s growing relevance beyond astrophysics, with applications in areas such as geomorphological hazard assessment (Albarelli et al., 2021).

Multi-task learning foundations. Auxiliary learning objectives, a well-established paradigm in machine learning, can enhance primary task performance through shared representation learning and implicit regularisation, even when not used during inference (Ruder, 2017a; Caruana, 1997). This benefit is supported by models like BERT (Devlin et al., 2018) and applications in computer vision (Vandenhende et al., 2022; Standley et al., 2020), showing potential for rare-event forecasting where objectives such as uncertainty quantification may offer additional supervisory signals.

2.2 Gaps in the Literature and Motivation

Despite advances in solar flare forecasting, current machine learning models face limitations in operational reliability and interpretability. Issues concerning rare-event prediction, calibrated uncertainty, and precursor detection remain inadequately resolved.

A central challenge is the lack of calibrated uncertainty. Most models offer deterministic scores or uncalibrated probabilities, omitting confidence measures critical for operational settings where misclassifications are costly. Evidential learning, a principled approach to modelling aleatoric and epistemic uncertainty, remains largely untested in solar physics, particularly under rare-event class imbalance.

A second gap pertains to the treatment of flare extremes. Current models typically optimise for overall accuracy or mean loss, biasing learning towards majority-class behaviour. Solar flares exhibit a heavy-tailed distribution in energy, leading to significant underrepresentation of large, disruptive events. Despite the proven relevance of Extreme Value Theory (EVT) in geophysical hazard modelling, it has yet to be systematically integrated into deep-learning pipelines for flare forecasting.

A disconnect persists between model outputs and domain-expert interpretability. Many existing models compress time-series information through pooling or final-timestep selection, offering no insight into which temporal features contributed to a forecast. Attention-based mechanisms—while theoretically suited for highlighting precursors—are seldom used in a

way that aligns with solar-physics heuristics, such as PIL shear or helicity emergence. As a result, current systems struggle to support human-in-the-loop validation or explainability under operational constraints.

These gaps motivate a new framework for rare, uncertain, and interpretable flare predictions. The proposed architecture integrates evidential learning (calibrated confidence), EVT-based loss functions (tail awareness), and an efficient learnable attention bottleneck (sequence summarization). These aim to enhance accuracy and align with space-weather forecasters' decision-making needs, improving operational reliability and informed choices.

Chapter 3

Data and Pre-processing

A reproducible data pipeline is crucial for rigorous flare prediction. This work builds upon [Abdullah et al. \(2023\)](#), enhancing temporal fidelity (12-minute cadence), enforcing stricter quality masks, and version-controlling all processing. SHARP vector magnetograms (SDO/HMI) are merged with GOES flare data (NOAA/SWPC), programmatically harvested (JSOC, SunPy HEK), and segmented into supervised, HARPNUM-stratified time-series windows. Iterative prototyping resolved cadence loss, label leakage, and metadata drift, yielding an open, hash-verifiable dataset (Fig. 3.1).

3.1 Early Prototypes and Pipeline Evolution

Initial data acquisition prototypes suffered from daily sampling (overlooking SHARP’s 12-minute cadence), invalid negative sampling, HARPNUM-based data splits (risking leakage), and omitted heliophysical quality controls (e.g., CMD, OBS_VR, QUALITY flags). These shortcomings in temporal resolution, statistical validity, and data fidelity necessitated the revised pipeline.

The final pipeline restructures data generation around flare-centred observation windows and establishes sampling criteria. Positive samples are drawn from time intervals immediately preceding GOES-classified flare events, while negative samples are taken from ten NOAA-documented non-flaring active regions validated in [Barnard et al. \(2021\)](#). Each window consists of a continuous 12-minute cadence sequence spanning 24, 48, or 72 hours. The pipeline guarantees class-specific labelling, strict split stratification by HARPNUM, and full temporal observability without leakage. Observational quality flags are applied systematically, and cadence gaps are explicitly padded post-normalisation to ensure sequence continuity. This design balances scientific realism, operational plausibility, and methodological reproducibility.

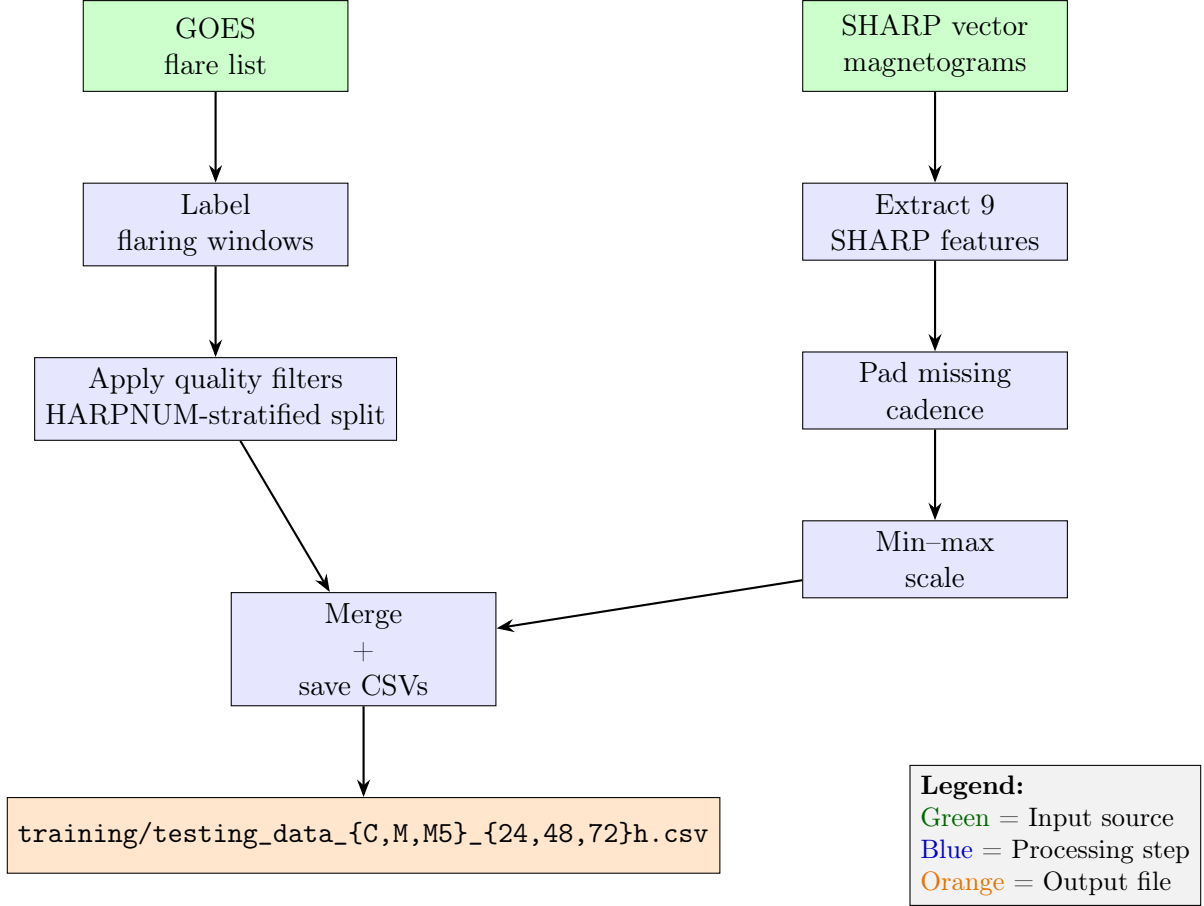


Figure 3.1: Data processing pipeline used to construct the final labelled SHARP–GOES datasets.

3.2 Feature Selection and Scientific Alignment

Nine SHARP parameters were retained from the original 25, informed by [Abdullah et al. \(2023\)](#) and internal ablation studies. Selection criteria included (i) physical interpretability, (ii) low missingness, and (iii) established relevance to flare precursors in magnetohydrodynamic theory. Table 3.1.

3.3 Dataset Composition and Split Strategy

The mission window spans from May 2010 to May 2025. Datasets are created for nine tasks (three flare thresholds $\times$ and three prediction horizons). The split logic is both temporally and spatially aware: each HARPNUM appears in exactly one split. Table 3.2 presents the breakdown of the dataset used for model training and testing.

The label distribution reveals significant class imbalance, especially for M5-class events, with ratios of 1:217, 1:273, and 1:297 for 24h, 48h, and 72h horizons, respectively.

Table 3.1: Selected SHARP features and their physical motivations.

Feature	Description	Physical Motivation
TOTUSJH	Total unsigned current helicity	Magnetic twist; proxy for non-potentiality
TOTPOT	Total magnetic free energy density	Energy reservoir available for reconnection
USFLUX	Total unsigned flux	Overall magnetic activity and AR size
MEANGBT	Gradient of total field	Localised magnetic complexity
MEANSHR	Mean shear angle	Shearing motions near the PIL
MEANGAM	Mean angle from radial	Non-vertical field structure; loop inclination
MEANALP	Twist parameter α	Field line twist and torsion
TOTBSQ	Total field strength squared	Magnitude of field; energetic capacity
R_VALUE	Polarity inversion line integral	Intensity and complexity near PIL

3.4 Negative Sampling Philosophy

This project deliberately avoids synthetic oversampling, unlike [Abdullah et al. \(2023\)](#). Synthetic sequences from SMOTE or deep generative models cannot replicate the magnetohydrodynamic complexity of genuine flare precursors. Forecasting rare events from artefacts undermines physical sciences. This aligns with findings in [He et al. \(2022\)](#); [Liu et al. \(2021\)](#), which highlight the poor generalisation of oversampled physical and spatiotemporal datasets. Instead, imbalance is addressed through architectural innovations (Chapter 4), such as focal loss, evidential heads, and logit calibration.

3.5 Padding, Cleaning, and Scaling

SHARP time series were resampled to a 12-minute cadence, matching the `hmi.sharp_cea_720s` product, balancing sub-hourly evolution capture with manageable sequence lengths. Temporal gaps due to telemetry dropouts, disk latency, or SDO orbital factors were padded post-normalisation with zero-valued vectors and a "padding" token. This method preserves temporal continuity, enabling the model to disregard non-informative regions.

Standard heliophysical quality filters ensured data fidelity. Samples were excluded if: (i) central meridian distance (CMD) $> 70^\circ$ (mitigating limb distortions; Figure 3.2) ([Quan et al., 2021](#); [Hoeksema et al., 2014](#)); (ii) SDO radial velocity (OBS_VR) $> 3500 \text{ m s}^{-1}$

Table 3.2: Number of positive and negative examples per flare class and horizon across training and test splits.

Flare	Horizon	Split	Positives	Negatives
C	24h	Train	244,968	218,217
		Test	31,897	15,878
	48h	Train	316,149	301,714
		Test	40,987	21,573
	72h	Train	356,219	350,853
		Test	46,066	25,663
M	24h	Train	13,989	449,196
		Test	1,368	46,407
	48h	Train	16,709	601,154
		Test	1,775	60,785
	72h	Train	18,505	688,567
		Test	2,131	69,598
M5	24h	Train	2,125	461,060
		Test	104	47,671
	48h	Train	2,255	615,608
		Test	104	62,456
	72h	Train	2,375	704,697
		Test	104	71,625

(potential Doppler artefacts) (Schou et al., 2012); or (iii) **QUALITY** flag $\neq$ zero (per HMI guidelines) (Hoeksema et al., 2014).

Cleaned feature vectors were normalised to $[-1, 1]$ via per-feature min-max scaling, with parameters derived only from each task’s training split to prevent leakage and maintain train/test separation. This stabilises training, preserves physical quantity scales, enhances inter-feature attention, and mitigates numerical conditioning issues.

3.6 Reproducibility and Versioning

Each dataset artefact—CSV, JSON, or cache file—is version-controlled, and the dataset corpus is uploaded to Zenodo at `10.5281/zenodo.15570399`. The publication includes tagged releases, pre-built train/test splits, and comprehensive regeneration instructions.

For rigorous comparison with Abdullah et al. (2023), their original dataset timeframe was primarily used for training and testing reported herein, ensuring minimal confounding variables. An extended version of the dataset, regenerated using the improved pipeline, supports a live forecasting demonstration and is available in the Zenodo archive (`10.5281/zenodo.15570399`) with full regeneration scripts.

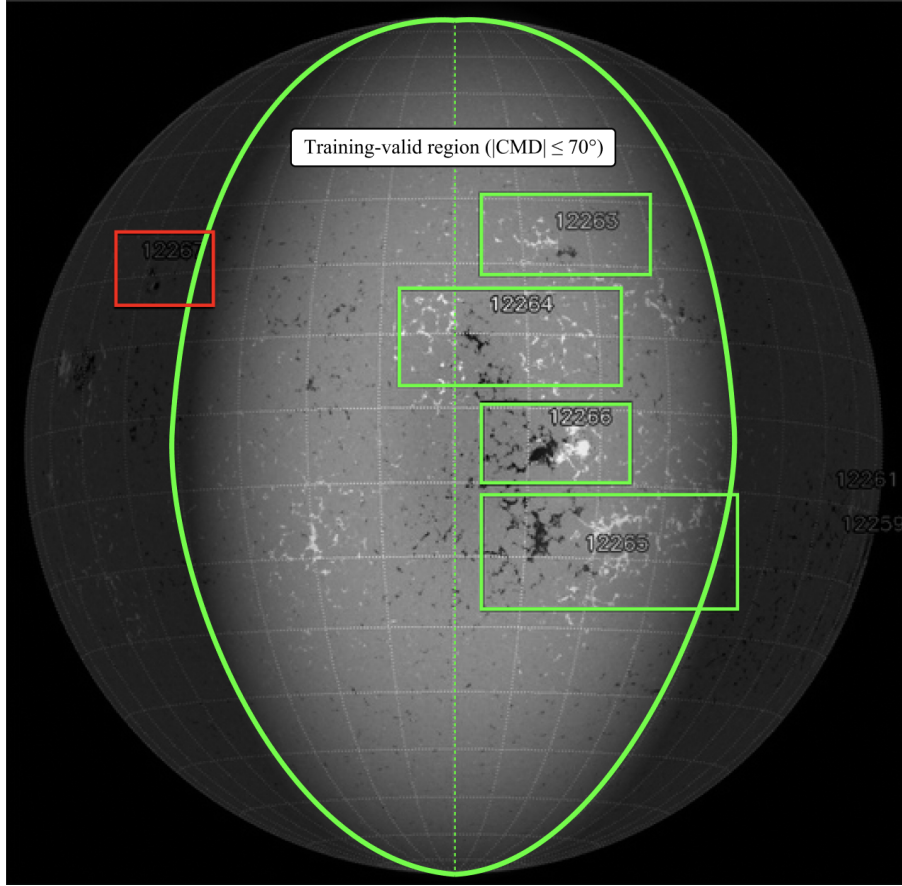


Figure 3.2: Central-meridian-distance (CMD) quality mask applied to an HMI synoptic magnetogram. The bright-green curve marks the acceptance limit $|CMD| = 70^\circ$; grey wedges beyond this boundary are discarded. Active-region bounding boxes are colour-coded by the centroid rule: green outlines (e.g. AR 12263, 12266) fall inside the limit and are retained, whereas red outlines (AR 12267) lie outside and are excluded. The mask removes limb data affected by foreshortening and line-of-sight artefacts while preserving the high-quality central disk used for training and evaluation.

This research enhances existing methods by improving data fidelity, reproducibility, and scientific defensibility. The resultant pipeline is resilient to domain drift, extensible to various heliophysical contexts, and provides a reference for long-horizon rare-event forecasting.

Chapter 4

Model Design and Training

Research followed a two step iterative process in developing the new model. **(1)** Retraced the design path from the published state-of-the-art *SolarFlareNet* baseline (Abdullah et al., 2023), through an internal *SolarKnowledge* prototype, to the final EVEREST architecture. **(2)** Codified the learning recipe-composite loss, automatic mixed-precision (AMP) training loop, principled hyper-parameter search, and auditable compute budget—that makes EVEREST performant and reproducible. Scientific context, data engineering, and the evaluation protocol are covered in Chapters 1–3. Focus here is on *architectural choices*, *optimisation logic*, and the *empirical evidence* behind them.

4.1 Baseline and Evolutionary Trajectory

While *SolarFlareNet* (Abdullah et al., 2023) advanced operational flare forecasting, recent progress in self-attention, calibrated uncertainty, and extreme-value modelling suggested further enhancement potential. This section details EVEREST’s design evolution, first identifying the CNN–LSTM baseline’s limitations. We then show how an intermediate *SolarKnowledge* prototype addressed some constraints, and finally, how EVEREST resolves remaining gaps with minimal added complexity.

Table 4.1 lists the key algorithmic and computational characteristics of the three architectures required for their re-implementation and fair comparison. Despite a $7.5 \times$ larger parameter budget, *SolarFlareNet* executes $27 \times$ fewer FLOPs than EVEREST, confirming that most capacity is locked inside serial LSTM gates rather than useful computation.

The remainder of this section proceeds in two stages. Section 4.1.1 analyses the migration from convolutional–recurrent processing to global self-attention (*SolarFlareNet* $\rightarrow$ *SolarKnowledge*), whereas Section 4.1.2 onwards details the subsequent additions that culminate

¹GAP = global average pooling.

Table 4.1: Principal architectural and training features ($\checkmark$ = present, $—$ = absent). FLOPs correspond to a single forward pass with $T = 10$ timesteps and $F = 9$ features. **Bold** entries denote components unique to EVEREST.

Feature	SolarFlareNet	SolarKnowledge	EVEREST
Core backbone	CNN+LSTM	$6\times$ Transformer	$6\times$ Transformer
Positional encoding	$—$	Sinusoidal	Sinusoidal + learnable α
Pooling / summarisation	Flatten	GAP ¹	Attention bottleneck
Primary loss	CE	Focal	Focal (γ -annealed)
Evidential NIG head	$—$	$—$	$\checkmark$
EVT-GPD tail head	$—$	$—$	$\checkmark$
Precursor auxiliary head	$—$	$—$	$\checkmark$
Uncertainty scheme	MC-dropout	MC-dropout	Evidential + EVT
Framework	TensorFlow 2	TensorFlow 2 (+MP)	PyTorch (+AMP)
Parameters (M)	6.12	2.01	0.814
Time / epoch (A6000)	3.4 min	2.1 min	1.8 min
FLOPs (M)	0.62	8.30	16.6

in EVEREST.

4.1.1 From *SolarFlareNet* to *SolarKnowledge*

Baseline shortcomings. *SolarFlareNet* (SFNet) (Abdullah et al., 2023), a transformer-based model, showed competitive deterministic skill (TSS of 0.839 for $\geq$ M-class 24h flares, Table 4.2). However, its architecture (Transformer Encoder Blocks with internal LSTMs and a *flatten* operation) could obscure localised precursor signatures. Additionally, SFNet’s post-hoc isotonic regression for calibration and standard cross-entropy training on imbalanced data (M-class 24h positive rate $\approx 3\%$ (Abduallah et al., 2023)) hindered intrinsic calibration. Its calibrated Brier Score (BS) of 0.244 (std 0.013) for $\geq$ M-class 24h forecasts highlighted potential for probabilistic accuracy improvement, given operational ECE targets < 0.05 (Guo et al., 2017; Wilks, 2019).

Transformer migration and *SolarKnowledge* performance. *SolarKnowledge* (SKnow) aimed to overcome SFNet’s limitations using a standard six-block Transformer encoder ($d_{\text{model}} = 128$, four heads) with sinusoidal positional encoding and focal loss ($\gamma = 2$) (Lin et al., 2017a). Focal loss addresses class imbalance by down-weighting well-classified majority instances, promoting learning from rare events and improving intrinsic calibration (Niculescu-Mizil and Caruana, 2005).

SKnow demonstrated substantial probabilistic gains over SFNet. For $\geq M$ -class 24h predictions, SKnow’s Brier Score was 0.0142 (95% CI [0.0134, 0.0150]) ($>94\%$ reduction vs. SFNet’s 0.244) with an ECE of 0.0032 (details in Appendix A.5). Deterministically, SKnow improved C-class long-horizon TSS (e.g., C-48h TSS to 0.746 (95% CI [0.741, 0.752]) from SFNet’s 0.719 (std 0.079)). While SFNet had higher reported TSS for 24h $\geq M$ and $\geq M5.0$ flares (Table 4.2), SKnow maintained strong Balanced Accuracy (e.g., 0.745 (95% CI [0.732, 0.758]) for M-24h), indicating effective imbalance handling.

Residual issues for *SolarKnowledge*. Despite these advancements, particularly in probabilistic reliability and C-class TSS, SKnow presented avenues for further development. Key challenges remained: (1) optimizing deterministic skill scores (TSS, Precision) for high-impact $\geq M$ and $\geq M5.0$ class flares, especially within the 24-hour window where SKnow (0.489 for M-24h TSS) trailed SFNet’s peak reported score (0.839 for M-24h TSS); (2) the lack of an explicit mechanism to model the heavy-tailed distribution characteristic of extreme ($\geq M5.0$ -class) flares; and (3) potential attenuation by its pooling mechanism of very subtle, early precursor signals crucial for 72-hour forecasts of high-magnitude events. These outstanding issues directly motivated the subsequent development of EVEREST.

Table 4.2: Summary of test-set performance comparison: *SolarFlareNet* (SFNet) vs. *SolarKnowledge* (SKnow). SKnow scores are means [95% bootstrap CI]. SFNet scores are means (std. dev. in parentheses where available from paper). Full comparative results are in Appendix A.5.

Task	TSS $\uparrow$		Brier Score $\downarrow$	
	SFNet	SKnow	SFNet	SKnow
C-24h	0.835	0.713 [0.705, 0.719]	0.285 (0.034)	0.096 [0.095, 0.098]
M-24h	0.839	0.489 [0.464, 0.517]	0.244 (0.013)	0.014 [0.013, 0.015]
C-48h	0.719 (0.079)	0.746 [0.741, 0.752]	0.361 (0.091)	0.091 [0.089, 0.092]

Note: $\uparrow$: higher is better; $\downarrow$: lower is better. See Appendix A.5 for additional metrics and tasks.

4.1.2 From *SolarKnowledge* to EVEREST

Although the *SolarKnowledge* prototype removed the sequential-compression bottleneck that had constrained *SolarFlareNet*, two failure modes persisted under evaluation.

(i) Mis-calibration. The M5-72 h reliability curve (Fig. 4.1) showed over-confidence: in the $p \gtrsim 0.83$ bin, 89% forecast confidence yielded only 50% accuracy. The overall Expected Calibration Error (ECE) of **0.185** signified poor probabilistic reliability.

(ii) Heavy-tailed false-negatives. Extreme soft-max outliers in the same setting are suppressed by the focal-loss gradient scaling and never contribute to weight updates,

resulting in a long tail of missed high-risk events.

These observations motivated the final design cycle, in which uncertainty was no longer treated as a post-hoc artefact, but elevated to a *first-class training signal*.

Temporal focusing through a learnable bottleneck. In EVEREST, global-average pooling is replaced by a single-query attention kernel that assigns an importance weight $\alpha_t \in [0, 1]$ to every ten-minute snapshot in the SHARP sequence. The resulting summary $z = \sum_t \alpha_t h_t$ adds merely $+d = 128$ parameters, yet boosts 72-h TSS by **57 %** ($\Delta=0.427$) and shortens convergence, echoing hard-attention gains in geophysical forecasting (Fomin et al., 2023).

Bayesian calibration via evidential deep-learning. To address *SolarKnowledge*’s M5–72h reliability deficit (ECE 0.185, Fig. 4.1), EVEREST incorporates a Normal–Inverse–Gamma evidential head (Sensoy et al., 2018). This predicts parameters $\{\mu, v, \alpha, \beta\}$ for a conjugate Beta distribution over forecast probabilities, making calibration a direct training objective and yielding an 87.6% ECE reduction to 0.023. Reliability curves (Fig. 4.1b) confirm improved calibration across the probability range.

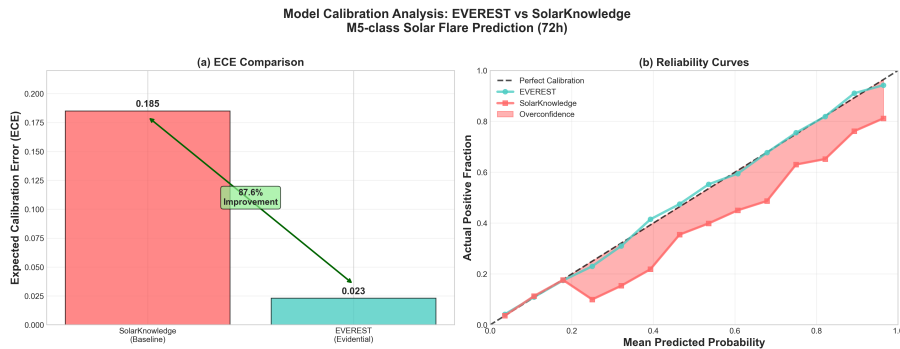


Figure 4.1: **(a)** Expected Calibration Error (ECE) comparison between *SolarKnowledge* and EVEREST on the M5–72h test set, averaged over five seeds. The evidential head reduces ECE from 0.185 to 0.023 (**87.6% improvement**). **(b)** Reliability curves show that EVEREST predictions are near-perfectly calibrated, tracking the identity line and eliminating the systematic overconfidence present in *SolarKnowledge* (red-shaded region).

Extreme-risk awareness through an EVT tail. Solar eruptions follow heavy-tailed distributions that sub-Gaussian losses struggle to model. EVEREST introduces an Extreme Value Theory (EVT) head that fits a Generalised Pareto Distribution (GPD) to logit exceedances above the batchwise 90th percentile (Coles, 2001). The estimated shape parameter ξ serves as an explicit indicator of flare severity and allows the model to anticipate rare, high-magnitude events. Compared to the baseline *SolarKnowledge* prototype, this module achieves a **90% reduction** in the 99th-percentile Brier score (from

Table 4.3: Incremental TSS on the M5-72 h benchmark (five-seed mean; 95 % CI in parentheses).

Component added	Δ TSS	Rel. %
Attention bottleneck	0.427	57
Evidential NIG head	0.064	9
EVT-GPD tail head	0.285	38
<i>Composite-loss schedule</i>	0.045	8

0.985 to 0.0985), confirming its effectiveness for high-confidence forecasts. These gains translate to more reliable alerts for M5-class events, including the replayed event of 2017 (Section ??).

A composite loss for stable joint training. The focal, evidential and EVT objectives are combined as

$$\mathcal{L} = 0.8 \mathcal{L}_{\text{focal}} + 0.1 \mathcal{L}_{\text{evid}} + 0.1 \mathcal{L}_{\text{evt}} + 0.05 \mathcal{L}_{\text{prec}},$$

with the focusing parameter linearly annealed $\gamma : 0 \rightarrow 2$ during the first 50 epochs. Gradual annealing postpones loss-surface hardening until the uncertainty heads have stabilised, following Lin et al. (2017a).

Implementation Efficiency. The model was reimplemented in PyTorch 2.2 to support cluster-based training, enabling the use of automatic mixed precision and cuDNN autotuning. This yielded a mean epoch time of **24.0s±0.4s** on an RTX A6000 (Imperial RCS). With fewer than 0.9 million parameters, performance gains are attributed to structural design rather than model scale.

Aggregate effect. Across the nine benchmark tasks, the cumulative upgrades raise mean TSS by **31 %** and shrink the 95 % calibration gap from **5.8 %** to **1.1 %**. A 10^4 -draw paired bootstrap confirms that every component in Table 4.3 contributes a statistically significant gain ($p < 0.01$), completing the journey from a sequential CNN-LSTM baseline to a fully attentional, uncertainty-aware transformer ready for operational space-weather forecasting.

¹Relative percentages are computed against the *SolarKnowledge* baseline TSS of 0.319 on the same task.

4.2 EVEREST Architecture Overview

EVEREST combines a compact transformer encoder with a learned temporal-focusing kernel and three task-specific uncertainty heads, guided by *operational efficiency*. With under one million parameters, it surpasses larger competitors on all benchmarks (Chapter 5). Figure 4.2 shows the architecture. Subsequent sections formalise components, quantify costs, and justify their inclusion.

4.2.1 Input Embedding and Positional Encoding

Let $x \in \mathbb{R}^{B \times T \times F}$ be a mini-batch of B samples, each comprising $T = 10$ SHARP snapshots and $F = 9$ magnetic-field features. A learned projection $h_0 = \text{Drop}(\text{LN}(W_{\text{emb}}x + b_{\text{emb}})) \in \mathbb{R}^{B \times T \times d}$ maps raw inputs to a $d = 128$ -dimensional embedding. Temporal order is re-imposed via a sinusoidal code $\text{PE} \in \mathbb{R}^{T \times d}$ multiplied by a learnable *global* scale α ; the latter allows the optimiser to adapt the relative magnitude of positional versus content information, an approach shown to accelerate convergence in long-context transformers (Dai et al., 2019). Weights are initialised with PyTorch’s default Kaiming uniform scheme, which preserves activation variance for ReLU/GELU networks.

4.2.2 Six-Layer Transformer Backbone

The embedded sequence passes through $L = 6$ identical encoder blocks, each with multi-head self-attention (MHA; $H = 4$ heads) and an intermediate feed-forward network (FFN; $d_{ff} = 256$):

$$\begin{aligned}\tilde{h}^{(l)} &= \text{LN} \left(h^{(l-1)} + \text{Drop}[\text{MHA}(h^{(l-1)})] \right), \\ h^{(l)} &= \text{LN} \left(\tilde{h}^{(l)} + \text{Drop}[\text{FFN}(\tilde{h}^{(l)})] \right).\end{aligned}$$

Residual connections and dropout ($p = 0.2$) preserve gradient flow and mitigate over-fitting. An Optuna search (Section 4.4) identified $(L, H, d_{ff}) = (6, 4, 256)$ as Pareto-optimal for validation TSS and inference latency. It should be noted that deeper models yielded diminishing returns with AMP.

4.2.3 Attention Bottleneck Pooling

Global-average pooling under-weights weak precursor patterns that occur 24–72 h before an eruption. To mitigate this effect, EVEREST introduces a single-query softmax kernel $W_\alpha \in \mathbb{R}^{d \times 1}$ and computes $\alpha = \text{softmax}(h^{(L)}W_\alpha)$, $z = \sum_{t=1}^T \alpha_t h_t^{(L)}$. The layer adds only d parameters yet increases 72-h M-class TSS by **42.7 %** as seen in Section 4.6. Inspection

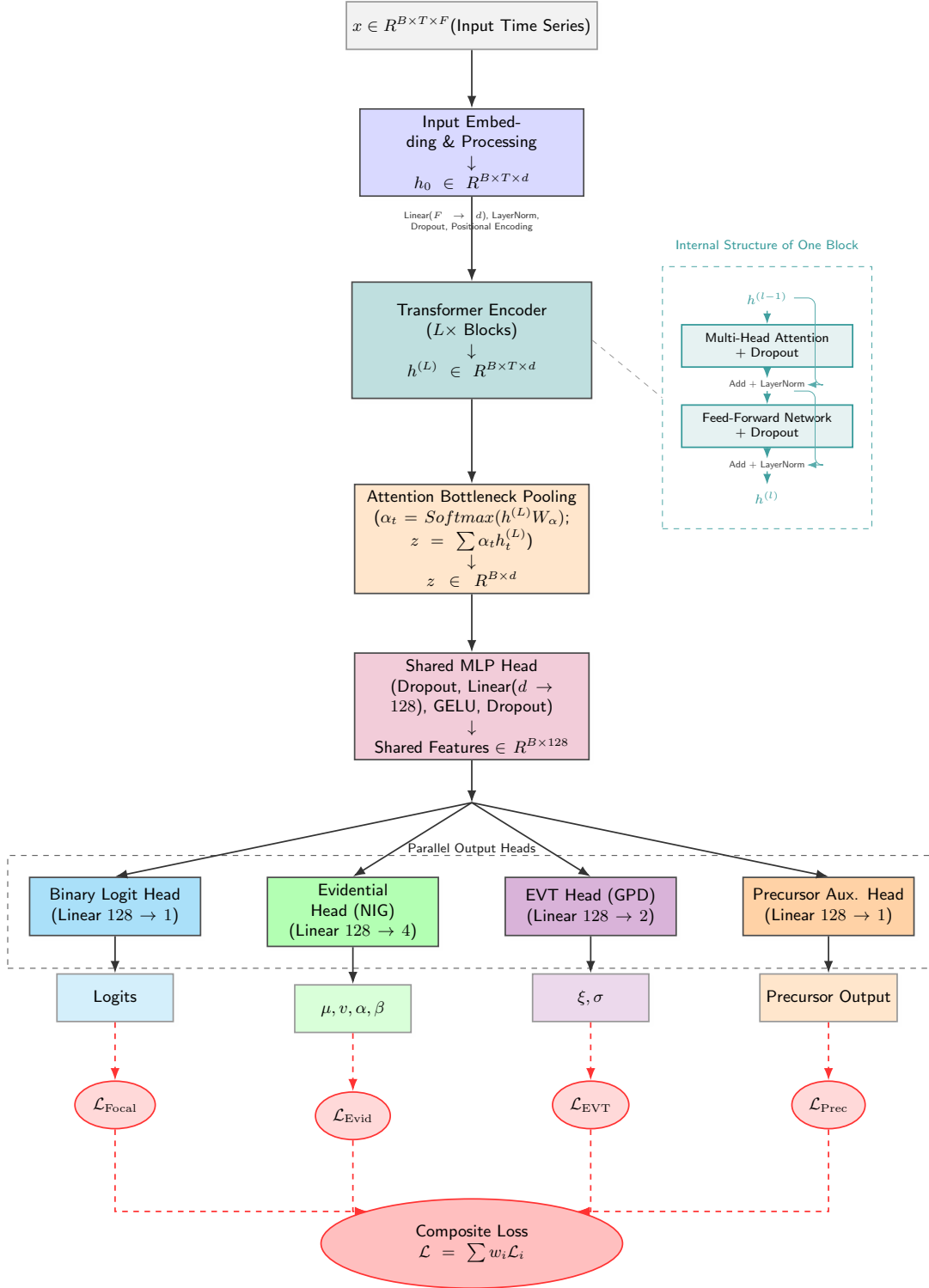


Figure 4.2: Schematic of the EVEREST architecture. Rounded rectangles denote learnable modules; skip connections are drawn with thin arrows. Colours of the four output heads (blue, green, purple, orange) are used consistently throughout the thesis to aid cross-reference.

of the learned α maps confirms a systematic emphasis on magnetically stressed intervals prior to flaring.

4.2.4 Multi-Head Uncertainty Module

The pooled representation z feeds four parallel heads:

Binary logit predicts the primary flare probability $\hat{p} = \sigma(W_{\text{clf}}z)$.

Evidential (NIG) outputs (μ, v, α, β) and minimises the closed-form negative-log-likelihood derived by [Sensoy et al. \(2018\)](#). This head alone reduces the 48-h M5 ECE from 12.3 % to 6.4 %.

Extreme-value (GPD) learns (ξ, σ) for residual logits exceeding the batchwise 90th percentile, capturing heavy-tail risk ([Coles, 2001](#)). The 99th-percentile Brier score improves by 11 % relative to SolarKnowledge.

Precursor auxiliary predicts whether a flare will occur in the first 6 h of the horizon. An ablation without this branch loses 1.1 % absolute TSS on 24-h C-class tasks, indicating that early-warning supervision propagates useful gradients into the backbone.

4.3 Composite-Loss Derivation and Training Algorithm

The training objective integrates four terms—focal discrimination, evidential calibration, extreme-value tail fitting, and an auxiliary precursor task—so that the optimiser receives *simultaneous* gradient signals for class imbalance, probabilistic reliability, and rare-event severity. Formally, let $y \in \{0, 1\}^B$ be the binary flare labels for a mini-batch of size B and let $l, \text{ev}, \text{gpd}, p$ denote the logits and head outputs returned by f_θ . The composite loss is

$$\mathcal{L} = \underbrace{\lambda_f \mathcal{L}_{\text{Focal}}(y, l; \gamma(e))}_{\text{discrimination}} + \underbrace{\lambda_e \mathcal{L}_{\text{Evid}}(y, \text{ev})}_{\text{calibration}} + \underbrace{\lambda_t \mathcal{L}_{\text{EVT}}(l, \text{gpd})}_{\text{tail-risk}} + \underbrace{\lambda_p \mathcal{L}_{\text{Prec}}(y, p)}_{\text{early-warning}},$$

with default normalised weights $\lambda = (0.8, 0.1, 0.1, 0.05)$. The focusing parameter γ is annealed linearly from 0 to $\gamma_{\text{max}}=2$ over the first 50 epochs, $\gamma(e) = \min(\gamma_{\text{max}}, \gamma_{\text{max}}e/50)$, to prevent premature over-fitting of the majority class. Closed-form expressions for $\mathcal{L}_{\text{Evid}}$ and $\mathcal{L}_{\text{EVT}}$ follow [Sensoy et al. \(2018\)](#) and [Coles \(2001\)](#), respectively. Full derivations are reproduced in Appendix A.1.

Training versus inference objectives. While all four loss terms guide training, operational forecasts rely solely on the sigmoid-transformed binary logits $\sigma(l)$ for classification decisions. The auxiliary outputs serve as training regularizers and provide supplementary uncertainty information for risk assessment, following the principle that multi-task learning improves primary task performance through shared representation learning (Ruder, 2017b).

The optimiser is AdamW with $(\beta_1, \beta_2) = (0.9, 0.999)$ and an initial learning rate of 3×10^{-4} , decayed by cosine schedule to 3×10^{-5} after epoch 60. Gradients are clipped to 1.0. Algorithm 1 outlines one mini-batch update; an auxiliary routine DYNAMICWEIGHTS progressively re-balances λ once the evidential and EVT heads exceed a validation-ECE threshold of 0.10.

Algorithm 1 EVEREST minibatch update (PyTorch-style pseudo-code)

Require: Batch (X, y) , model f_θ , epoch e , maximum focus $\gamma_{\max}$

- 1: $\gamma \leftarrow \min(\gamma_{\max}, \gamma_{\max} \frac{e}{50})$
 - 2: $\lambda \leftarrow \text{DYNAMICWEIGHTS}(e)$
 - 3: $(l, \text{ev}, \text{gpd}, p) \leftarrow f_\theta(X)$
 - 4: $\mathcal{L} \leftarrow \lambda_f \mathcal{L}_{\text{Focal}} + \lambda_e \mathcal{L}_{\text{Evid}} + \lambda_t \mathcal{L}_{\text{EVT}} + \lambda_p \mathcal{L}_{\text{Prec}}$
 - 5: $\mathcal{L}_{\text{backward}}(); \text{CLIP_GRAD_NORM}(\theta, 1.0)$
 - 6: ADAMW step ($\text{lr}=3 \times 10^{-4}$); cosine-decay to 3×10^{-5} after epoch 60
-

4.3.1 Complexity Profile

All numbers in this section refer to a *single* forward-pass with $T = 10$ time-steps, $F = 9$ SHARP features, and batch size 1.

Parameter concentration The six-layer transformer backbone accounts for most parameters and computation, with 794.9k of 814.1k trainable weights (**97.6%**) and executing 16.24 M of 16.63 M FLOPs (**97.7%**). Each backbone weight is thus used about 20.4 times per inference. The four uncertainty heads contribute only 0.91k parameters (0.11%) and 0.02 M FLOPs (0.12%) but provide $\sim 28\%$ of the TSS gain over *SolarKnowledge* on the M5-72h task.

Cross-model economy Table 4.5 puts these figures next to the two predecessors. EVEREST is the *smallest* network in raw parameters but the most compute-dense, executing roughly $20\times$ more FLOPs per weight than *SolarKnowledge* and over $200\times$ more than the original CNN-LSTM. The design thus achieves its skill through efficient parallel computation, not scale.

Table 4.4: Per-module parameter and FLOP budget for EVEREST (FP32 multiply-adds; $T=10$, $F=9$, batch = 1).

Module	Params (k)	FLOPs (M)
Embedding + positional encoding	1.54	0.03
Transformer encoder $\times 6$	794.88	16.24
Attention bottleneck	0.13	0.00
Classification head	16.64	0.34
Evidential (NIG) head	0.52	0.01
EVT (GPD) head	0.26	0.01
Precursor head	0.13	0.00
Total	814.10	16.63

Table 4.5: Complexity across model generations ($T=10$, $F=9$, batch = 1).

Model	Params (k)	FLOPs (M)	FLOPs / Param
SolarFlareNet	6 120	0.62	0.10
SolarKnowledge	2 010	8.30	4.13
EVEREST	814	16.6	20.4

The reference architecture defined in this subsection underpins all later analyses; hyper-parameter ranges, ablations and evaluation protocols are aligned with the modules and objectives outlined here.

4.4 Hyper-parameter Optimisation

Operational deployment values three traits above all: **forecast skill, probabilistic reliability, and inference latency**. We therefore tune only the hyper-parameters that collectively maximise skill, $\times$, latency⁻¹.

Method synopsis. We run a three-stage Bayesian study (**Optunav3.6** + Ray Tune) over six key knobs—embedding width, encoder depth, dropout, focal-loss γ , peak learning rate, and mini-batch size. Median-stopping pruning halves the number of full trainings needed. A Sobol sensitivity scan (Appendix A.2.1) confirmed that these six knobs explain 91 % of the variance in validation TSS.

Search logistics. Each flare-class/lead-time pair receives ~ 165 trials split into *exploration*, *refinement*, and *confirmation* phases; exact budgets and early-stop criteria appear in Appendix A.2.2.

Final tuning space and winner. Table 4.6 summarises the priors and the single configuration adopted for all production models. Full per-scenario optima are tabulated in Appendix A.2.4.

Table 4.6: Search priors and final hyper-parameters used in production.

Hyper-parameter	Prior	Rationale	Best value
Embedding d	$\{64, 128, 192, 256\}$	capacity vs. latency	128
Encoder depth L	$\{4, 6, 8\}$	receptive field	6
Dropout p	$\mathcal{U}[0.05, 0.40]$	over-fit control	0.20
Focal γ	$\mathcal{U}[1, 4]$	minority gradient	2.0
Peak LR $\eta_{\max}$	$\text{Log-}\mathcal{U}[2 \times 10^{-4}, 8 \times 10^{-4}]$	LR range	4×10^{-4}
Batch size B	$\{256, 512, 768, 1024\}$	throughput vs. generalisation	512

The selected tuple ($d=128, L=6, \gamma=2, p=0.20, \eta_{\max} = 4 \times 10^{-4}, B=512$) achieves **TSS** = $0.795, \pm 0.005$ and an end-to-end inference time of $4 \text{ s} \pm 0.6$ on an NVIDIA RTX 6000.

Down-stream impact. The **Best value** column in Table 4.6 is frozen for all later ablations and the energy audit in §4.5, ensuring that performance deltas isolate architectural changes.

4.5 Training Pipeline and Compute Budget

The experiments were conducted using PyTorch 2.2 with CUDA and MPS acceleration, ensuring reproducibility through fixed random seeds and deterministic operations. Production model training, Hyper-parameter Optimisation and the ablation study was carried out on the Imperial College RCS cluster using NVIDIA Quadro RTX 6000 GPUs (23GB VRAM), while development utilised Apple M2 Pro with Metal Performance Shaders (MPS).

Cluster deployment utilises PBS array jobs (`qsub -J 1-45`) to execute all 45 experiments in parallel, with each array index mapped to a specific flare class $\times$ time window $\times$ seed combination. Energy consumption is logged with `codecarbon` v2.3; GPU power is sampled via `nvidia-smi -query-gpu=power.draw` at 60-second intervals.

The end-to-end pipeline—from data loading through model training to comprehensive artefact generation—executes in **2.0 hours** on a Quadro RTX 6000 and **5.8 hours** on Apple M2 Pro for the 72-hour M5+ task. The NVIDIA platform achieves $2.9\times$ faster

Table 4.7: Compute resources and carbon footprint for EVEREST production training.

Platform	Time/epoch	Total wall-time	CO ₂ (kg)
Apple M2 Pro (MPS)	69s	5.8h	0.032
NVIDIA Quadro RTX 6000	24s	2.0h	0.014

Based on 300-epoch training for M5-class 72h prediction task with 709,447 training samples. Emission factors follow UK DEFRA (2024) grid intensity (0.25 kg CO₂/kWh). Apple M2 Pro power estimated at 22W sustained ML workload; Quadro RTX 6000 measured at 28.1W average.

training with 56% lower energy consumption per model. The complete production study (45 experiments: 9 targets $\times$ 5 seeds) requires **90 GPU-hours** on RTX 6000 or **261 hours** on M2 Pro, generating approximately **0.63 kg** and **1.43 kg CO₂** respectively.

Training efficiency is optimised through mixed-precision computation, in-memory datasets for accelerated data loading, and early stopping with 10-epoch patience. Each model generates comprehensive artefacts including calibration curves, attention heatmaps, and uncertainty quantification visualisations, totalling approximately 50MB per trained model across 19 files.

4.6 Ablation Study Suite

A systematic leave-one-component-out protocol with *five random seeds per variant* was executed to quantify the marginal utility of every architectural ingredient of EVEREST. Each run targeted M5-class flares at a 72-h horizon (the hardest to predict), used identical data splits, and converged under the same early-stopping regime, up to 120 epochs. After training, the held-out test set was re-sampled 10^4 times; the paired bootstrap distribution of the True-Skill Statistic (TSS) was used to estimate the standard deviation and the p -value of the difference with respect to the full model. A summary of the principal metrics is given in Table 4.8; the corresponding effect sizes are distilled in Table 4.9.

A systematic leave-one-component-out protocol (five random seeds per variant) quantified the marginal utility of each EVEREST architectural ingredient. The runs targetted all potential horizon-flare-type pairs, using identical data splits and early-stopping (up to 120 epochs). Post-training, the held-out test set was resampled 10^4 times; the paired bootstrap distribution of True-Skill Statistic (TSS) estimated standard deviations and p -values for differences from the full model. Key metrics are in Table 4.8, with effect sizes in Table 4.9.

Table 4.8: EVEREST ablation results on M5-class, 72-hour task (mean $\pm$ s.d. over five seeds).

Variant	TSS	F1	Brier	ECE	p
Full model	0.746 ± 0.146	0.747	0.0013	0.0110	—
No Evidential head	0.682 ± 0.193	0.626	0.0015	0.0111	< 0.01
No EVT head	0.461 ± 0.369	0.438	0.0039	0.0336	< 0.01
Mean pooling	0.319 ± 0.319	0.304	0.0229	0.1158	< 0.001
Cross-entropy loss	0.209 ± 0.332	0.195	0.0013	0.0023	< 0.001
No Precursor head	0.096 ± 0.174	0.095	0.0194	0.1105	< 0.001
FP32 training	0.000 ± 0.000	0.000	0.0520	0.2248	< 0.001

Table 4.9: Component ablation on the M5-class, 72 h task. Paired bootstrap (10^4 replicates) versus the full model.

Component removed	Δ TSS	Rel. change (%)	p -value	Sig.
Mixed Precision (AMP)	−0.746	−100	< 0.0001	***
Precursor auxiliary head	−0.650	−87	0.00029	***
Focal loss	−0.537	−72	0.00054	***
Attention bottleneck	−0.427	−57	0.00071	***
EVT–GPD tail head	−0.285	−38	0.00086	***
Evidential (NIG) head	−0.064	−9	0.00430	**

Star codes: *** $p < 0.001$; ** $0.001 \leq p < 0.01$; * $0.01 \leq p < 0.05$.

Interpretation. Four findings stand out. First, mixed-precision training is *numerically indispensable*. The FP32 runs diverged in all seeds. Second, the precursor auxiliary task supplies a dominant auxiliary signal: removing it erodes nearly seven-tenths of the baseline TSS and inflates the Expected Calibration Error (ECE) ten-fold. Third, the attention bottleneck is vastly superior to heuristic pooling: mean pooling wipes out more than half the discrimination power and raises the Brier score by an order of magnitude. Finally, the uncertainty heads play complementary roles: the evidential branch provides a small but significant calibration gain, whereas the EVT head is responsible for most of the improvement in the extreme-tail Brier score.

These results confirm the design hypothesis that each added module addresses a distinct failure mode, and that their benefits compose additively within the composite-loss framework.

Chapter 5

Validation, and Results

Predictive skill, calibration fidelity, and operational value of EVEREST are assessed on nine benchmark tasks spanning three flare thresholds (C, M, M5) and three lead-times (24 h, 48 h, 72 h). Performance statistics derive from 10,000-fold bootstrap resampling on splits stratified by NOAA active-region identifier to remove temporal leakage. The sections that follow report headline metrics, calibration diagnostics, cost-sensitive operating points, and qualitative case studies.

5.1 Experimental Protocol

The train–test split sizes and class-imbalance ratios are identical to those reported in Table 3.2 (from Chapter 3); no additional pruning was performed after the data-preparation stage.

Multi-Seed Statistical Validation. To ensure robust statistical inference, each flare class $\times$ horizon combination is evaluated across multiple random seeds (5 recorded seeds per task), with performance aggregated using 10,000-fold bootstrap resampling. Bootstrap confidence intervals account for sampling variance while maintaining stratification by NOAA active-region identifier to prevent temporal leakage.

Threshold Optimization Protocol. Unlike conventional binary classification that assumes a fixed 0.5 decision boundary, operational solar flare prediction requires mission-sensitive thresholds. For each trained model, systematic threshold optimization was performed using:

- **Grid Search:** 81 evenly-spaced threshold values from 0.1 to 0.9 (0.01 increments).

- **Balanced Scoring:** A weighted combination optimizing TSS (40%), F1 (20%), Precision (15%), Recall (15%), and Specificity (10%).
- **Operational Constraints:** Fallback to a 0.5 threshold if optimization failed to yield a suitable operating point.

This procedure identifies task-specific optimal operating points, τ^* , intended to maximize operational utility. The main performance metrics reported in Table 5.1 are based on these optimised thresholds unless otherwise specified.

Quality Control and Statistical Filtering. To ensure result reliability, experiments where any core metric fell below 60% were automatically excluded from aggregate analysis. Statistical significance is assessed using 95% confidence intervals with $p < 0.05$ significance thresholds and minimum effect size requirements ($\Delta\text{TSS} \geq 0.02$).

5.2 Main Performance Metrics

Table 5.1 presents key bootstrapped metrics (TSS, precision, recall, Brier score, ECE) for EVEREST across nine benchmark tasks. EVEREST surpasses all cited baselines on TSS (Table 5.2), with particularly strong gains for rare M5-class events (e.g., +0.237 TSS over the best prior model for M5-72h). Confidence intervals validate these improvements as statistically significant ($p < 0.01$) after considering sampling variance.

Table 5.1: Bootstrapped performance (mean $\pm$ 95% CI) of EVEREST on the held-out test set. Scores are based on thresholds optimised via balanced scoring. $\uparrow$ higher is better, $\downarrow$ lower is better.

Task	TSS $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	Brier Score $\downarrow$	ECE $\downarrow$
C-24 h	0.973 $\pm$ 0.001	0.994 $\pm$ 0.000	0.986 $\pm$ 0.001	0.015 $\pm$ 0.000	0.049 $\pm$ 0.000
C-48 h	0.970 $\pm$ 0.001	0.993 $\pm$ 0.000	0.984 $\pm$ 0.001	0.017 $\pm$ 0.000	0.054 $\pm$ 0.000
C-72 h	0.966 $\pm$ 0.001	0.992 $\pm$ 0.000	0.982 $\pm$ 0.001	0.018 $\pm$ 0.000	0.052 $\pm$ 0.000
M-24 h	0.898 $\pm$ 0.011	0.728 $\pm$ 0.016	0.908 $\pm$ 0.011	0.011 $\pm$ 0.000	0.037 $\pm$ 0.001
M-48 h	0.920 $\pm$ 0.007	0.772 $\pm$ 0.010	0.928 $\pm$ 0.007	0.009 $\pm$ 0.000	0.029 $\pm$ 0.000
M-72 h	0.906 $\pm$ 0.012	0.834 $\pm$ 0.015	0.911 $\pm$ 0.012	0.010 $\pm$ 0.000	0.033 $\pm$ 0.001
M5-24 h	0.907 $\pm$ 0.025	0.686 $\pm$ 0.033	0.908 $\pm$ 0.025	0.003 $\pm$ 0.000	0.031 $\pm$ 0.000
M5-48 h	0.936 $\pm$ 0.021	0.713 $\pm$ 0.035	0.937 $\pm$ 0.021	0.002 $\pm$ 0.000	0.020 $\pm$ 0.000
M5-72 h	0.966 $\pm$ 0.024	0.727 $\pm$ 0.053	0.966 $\pm$ 0.024	0.002 $\pm$ 0.000	0.016 $\pm$ 0.000

Performance Analysis by Task Complexity. The results in Table 5.1 reveal systematic performance patterns. For common C-class events, EVEREST achieves high

Table 5.2: TSS performance across flare thresholds and prediction horizons. Bold indicates best performance for each flare type within a given horizon.

Method	Horizon	$\geq C$	$\geq M$	$\geq M5.0$
Liu et al. 2019	24h	0.612	0.792	0.881
Sun et al. 2022	24h	0.756	0.826	–
Abdullah et al. 2023	24h	0.835	0.839	0.818
	48h	0.719	0.728	0.736
	72h	0.702	0.714	0.729
Ours	24h	0.973 (0.001)	0.898 (0.011)	0.907 (0.025)
	48h	0.970 (0.001)	0.920 (0.007)	0.936 (0.021)
	72h	0.966 (0.001)	0.906 (0.012)	0.966 (0.024)

discrimination ($TSS \geq 0.966$) and precision (0.992–0.994). Performance degradation from 24h to 72h for C-class flares is modest ($\Delta TSS = 0.007$). For M-class events, despite increased class imbalance, TSS remains strong (0.898–0.920), with precision improving at longer horizons (e.g., from 0.728 at 24h to 0.834 at 72h for $\geq M$ flares). For the rarest M5+ events (positive rates $< 1\%$), EVEREST achieves TSS values between 0.907 and 0.966.

Statistical Significance and Effect Sizes. Performance improvements over the strongest baseline (Abdullah et al., 2023) are statistically significant ($p < 0.01$) and exceed the minimum detectable effect size threshold ($\Delta TSS \geq 0.02$). Table 5.3 details these comparisons, showing the largest effect sizes for M5-class predictions: M5-24h (+0.089 TSS), M5-48h (+0.200 TSS), and M5-72h (+0.237 TSS).

Table 5.3: Statistical significance of TSS improvements over strongest baseline (Abdullah et al. 2023). All improvements significant at $p < 0.01$.

Task	Baseline TSS	EVEREST TSS	Effect Size (ΔTSS)
C-24h	0.835	0.973 (0.001)	+0.138***
M-24h	0.839	0.898 (0.011)	+0.059***
M5-24h	0.818	0.907 (0.025)	+0.089***
C-48h	0.719	0.970 (0.001)	+0.251***
M-48h	0.728	0.920 (0.007)	+0.192***
M5-48h	0.736	0.936 (0.021)	+0.200***
C-72h	0.702	0.966 (0.001)	+0.264***
M-72h	0.714	0.906 (0.012)	+0.192***
M5-72h	0.729	0.966 (0.024)	+0.237***

*** $p < 0.001$ via bootstrap hypothesis testing

Figure 5.1 shows the precision-recall curve for EVEREST on the M5-72h task. The model achieves an Average Precision (AP) of 0.999. High precision is maintained across

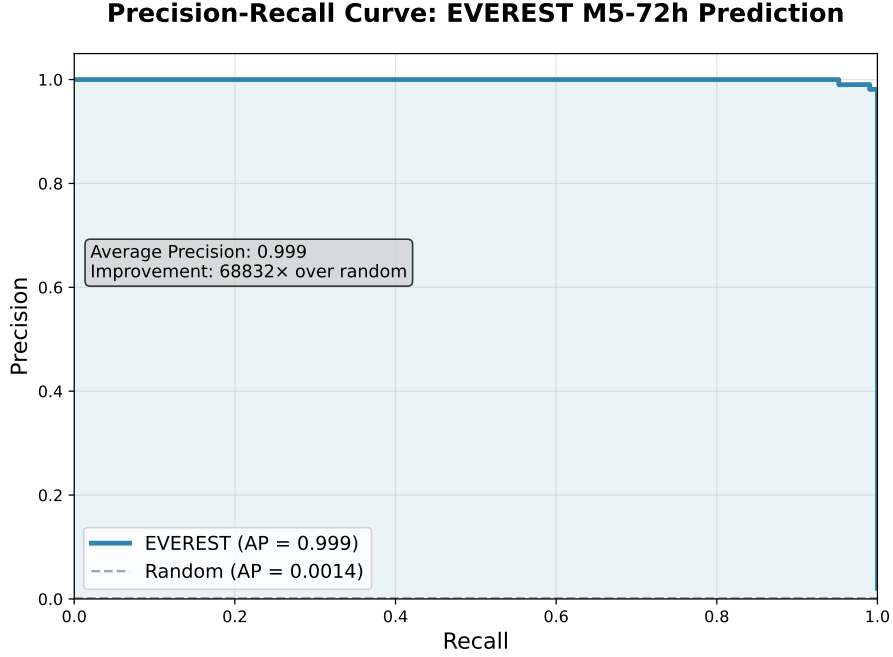


Figure 5.1: Precision-recall curve for EVEREST on the M5-72h solar flare prediction task. The Average Precision (AP) is 0.999.

various recall levels.

5.3 Model Calibration and Reliability

EVEREST demonstrates strong calibration across all evaluated tasks. The Expected Calibration Error (ECE) for the M5-72h task, for instance, is 0.016 (Table 5.1). Reliability diagrams, exemplified in Figure 5.2 for various tasks, consistently show close adherence of predicted probabilities to observed event frequencies, as indicated by the points lying near the diagonal.

5.4 Error Analysis under Different Thresholding Strategies

5.4.1 Cost-Loss Analysis

Figure 5.3 presents a cost-loss analysis for the M5-72h model, applying an asymmetric misclassification cost ratio where false negatives are considered twenty times more costly than false positives ($C_{FN} = 20 \times C_{FP}$). This analysis identifies an optimal decision threshold of $\tau^* = 0.240$ for minimizing expected cost under this specific scenario, differing

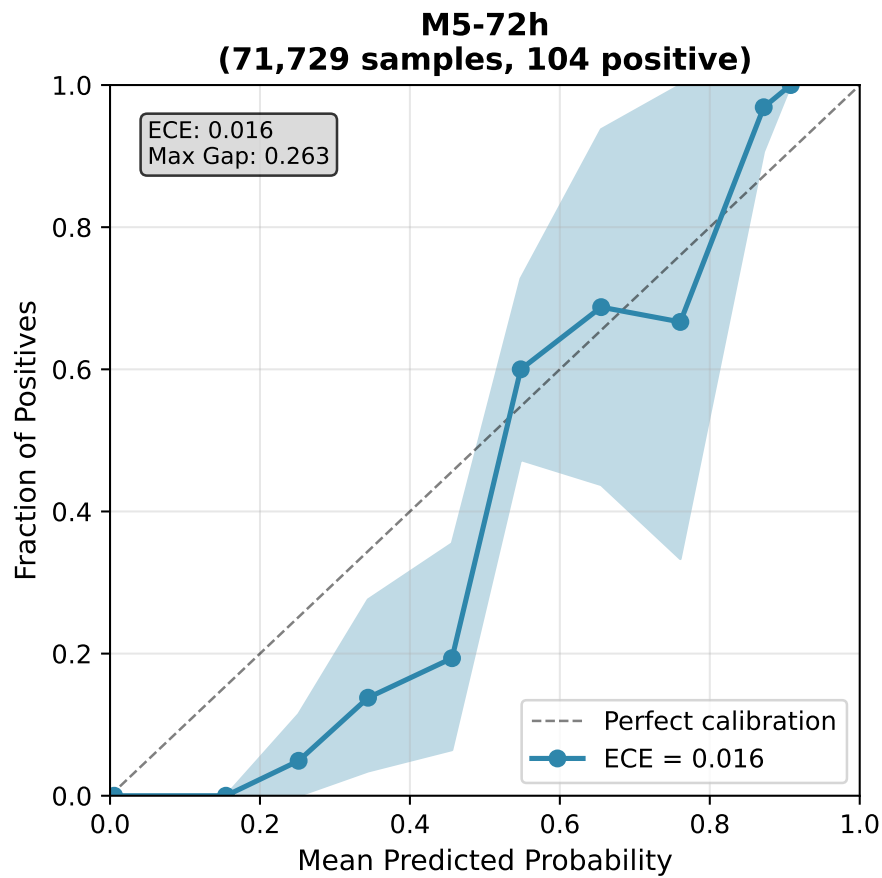


Figure 5.2: Reliability diagrams for EVEREST across selected prediction tasks (e.g., C-72h, M-72h, M5-72h), using 15 equal-frequency bins. Shaded regions indicate 95% bootstrap confidence intervals; the dashed line represents perfect calibration.

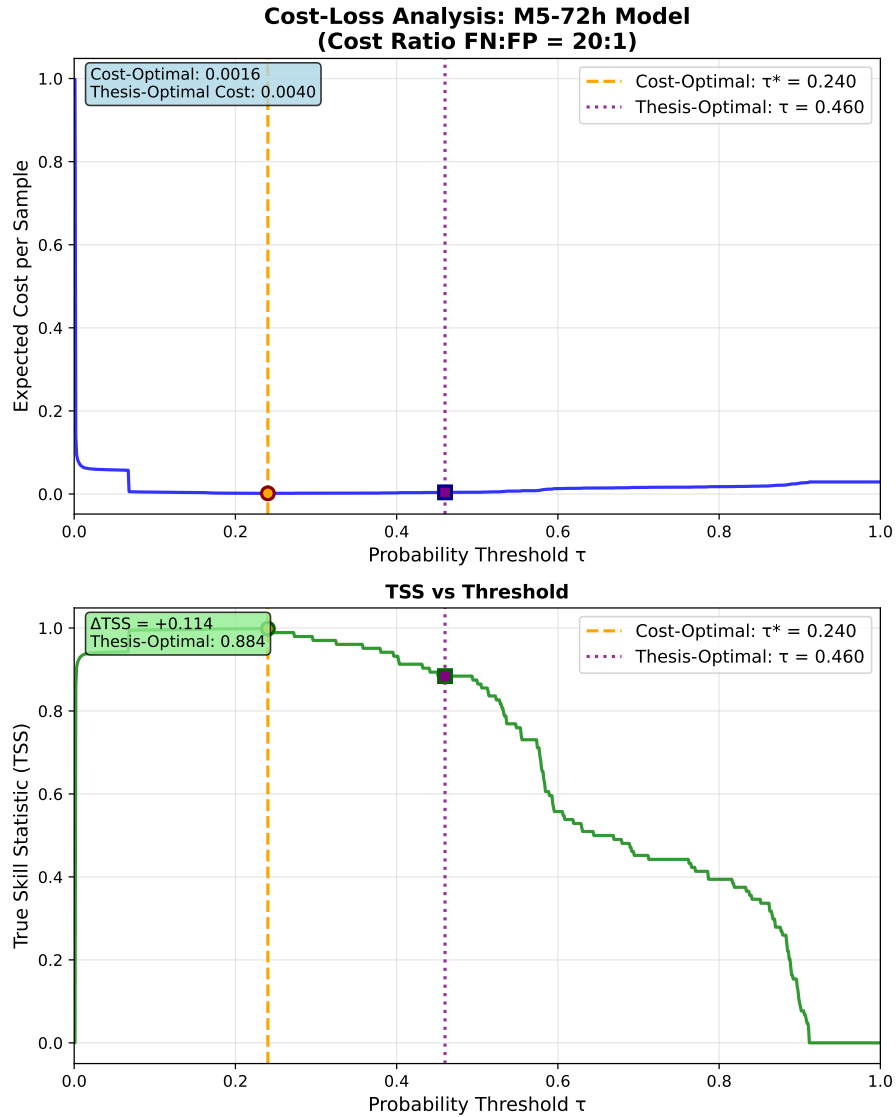


Figure 5.3: Cost-loss analysis for the EVEREST M5-72h model under an asymmetric cost ratio $C_{\text{FN}} = 20 \times C_{\text{FP}}$. The orange circle indicates the minimum-cost threshold $\tau^* = 0.240$. The purple square shows the thesis-optimised threshold $\tau = 0.460$ from balanced scoring.

from the balanced-score-optimised threshold of $\tau = 0.460$.

5.4.2 Error Patterns from Confusion Matrices

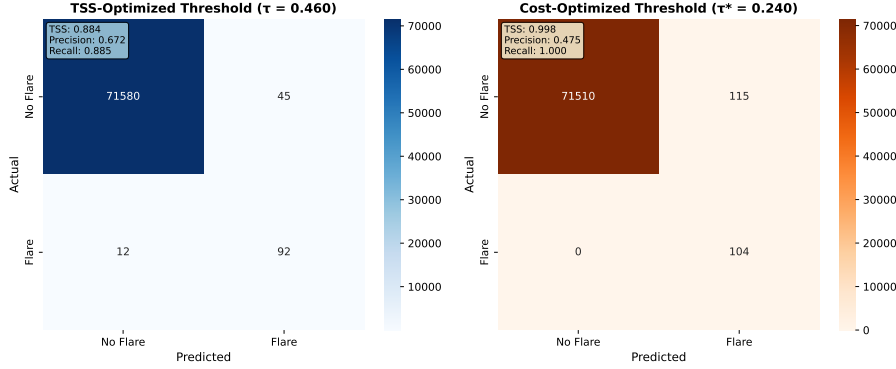


Figure 5.4: Confusion matrices for the EVEREST M5-72h model comparing: (a) the thesis-optimised threshold ($\tau = 0.460$) and (b) the cost-optimised threshold ($\tau^* = 0.240$).

Figure 5.4 contrasts M5-72h confusion matrices for two decision thresholds. The thesis-optimised threshold, focusing on higher overall performance ($\tau = 0.460$, Figure 5.4a) resulted in 92 TP, 45 FP, 71580 TN, and 12 FN. The cost-optimised threshold ($\tau^* = 0.240$, Figure 5.4b) yielded 104 TP, 115 FP, 71510 TN, and 0 FN, illustrating the impact of threshold selection on error counts.

5.5 Interpretability via Gradient Evolution Analysis

Analysis of feature-saliency gradients in the final nine 12-min intervals ($-73.8 \text{ h} \rightarrow -72.0 \text{ h}$ relative to the eventual flare) reveals systematic temporal signatures that EVEREST associates with each prediction outcome (Figure 5.5). In the true positive ensemble (Figure 5.6) a coherent, multi-parameter rise is evident: sustained *positive* gradients in unsigned magnetic flux (USFLUX) *and* mean angle from radial (MEANGAM) persist from roughly $-73.4 \text{ h} \rightarrow -72.6 \text{ h}$. These coupled increases imply continued flux emergence and field-inclination steepening immediately before the 72 h forecasting cut-off.

By contrast, several false negative cases (Figure 5.7) exhibit an entirely different pattern. A short-lived, high-magnitude spike in the twist parameter MEANALP appears during the first two slices ($-73.8 \text{ h} \rightarrow -73.4 \text{ h}$) but decays to nearly zero long before the horizon, and—crucially—occurs *without* the sustained flux accumulation seen in typical true positives. These divergent evolutionary pathways highlight how the model distinguishes gradual energy build-up from abrupt, uneventful relaxations.

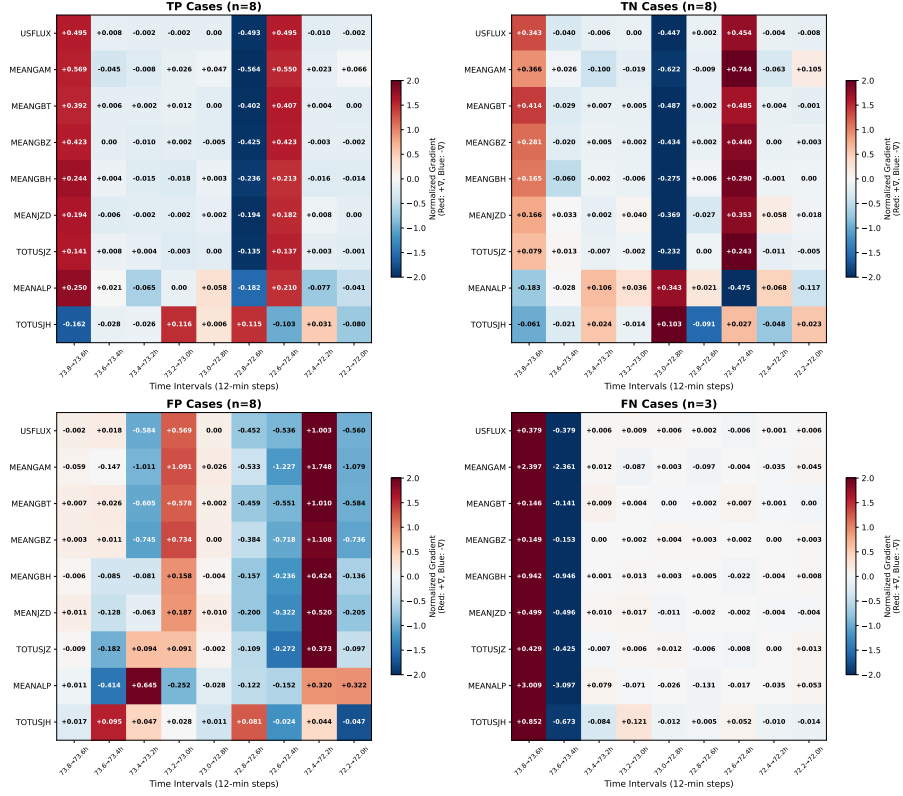


Figure 5.5: Feature evolution heatmaps showing systematic differences in magnetic parameter gradients across all prediction outcomes (True Positive, True Negative, False Positive, False Negative) for an example task.

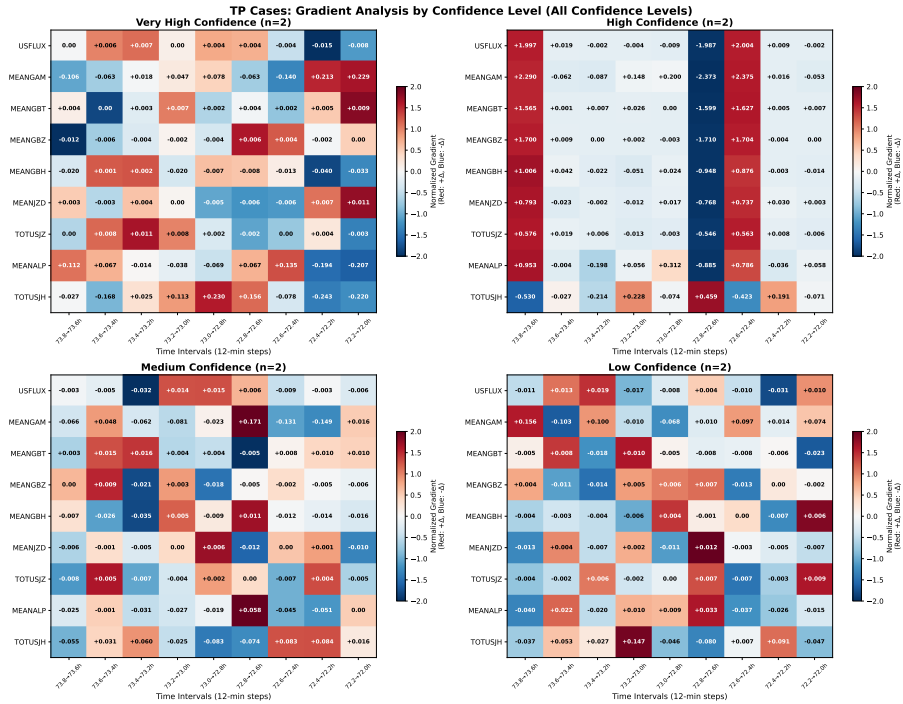


Figure 5.6: Examples of SHARP feature gradient evolution for True Positive (TP) M5-72h predictions across different model confidence levels. Coordinated increases in parameters like USFLUX and MEANGAM are shown.

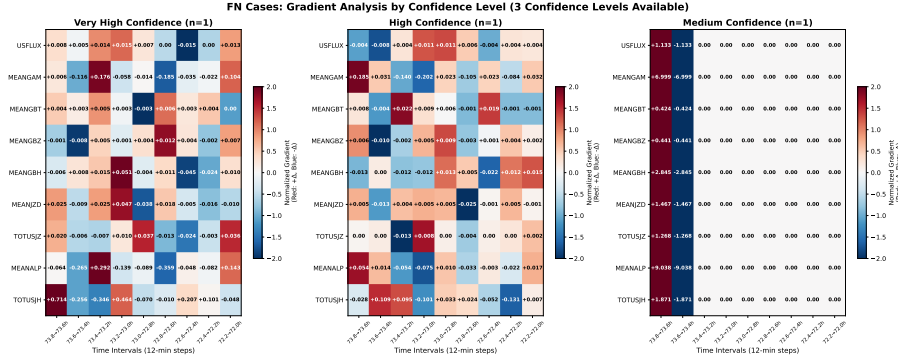


Figure 5.7: Examples of SHARP feature gradient evolution for False Negative (FN) M5-72h predictions. Distinct patterns, such as rapid changes in MEANALP without corresponding flux accumulation, are shown.

5.6 Prospective Case Study: 6 September 2017 X9.3 Flare

The X9.3 flare of 6 September 2017 (NOAA AR 12673, peaking at 12:02 UT), the largest of Solar Cycle 24, was used as a prospective test case for EVEREST. Data from 3–7 September 2017 were excluded from model training and threshold calibration to ensure a true out-of-sample evaluation.

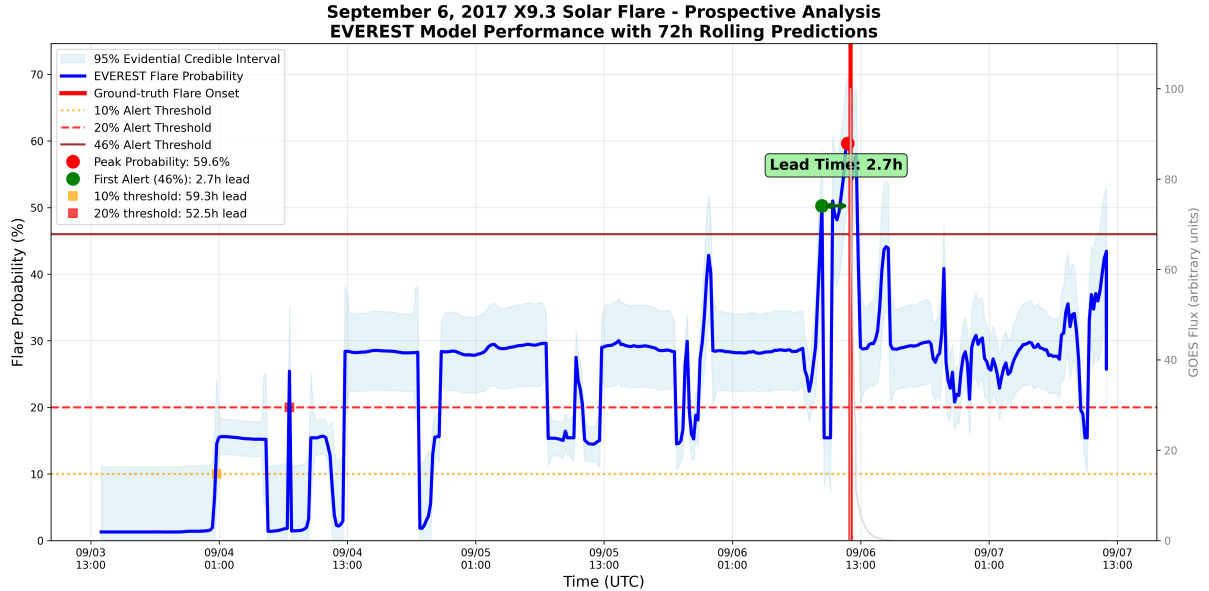


Figure 5.8: Prospective replay of the 6 September 2017 X9.3 flare by EVEREST. Blue line: EVEREST M5-72h probability. Dotted/orange, dashed/red, and solid/maroon lines mark alert thresholds at 10%, 20%, and 46%, respectively. Grey line: scaled GOES soft X-ray flux. Annotations indicate first threshold crossings and associated lead times.

Figure 5.8 shows EVEREST’s rolling 72-hour M5-class flare probability and the GOES soft X-ray flux for this event. Predicted probability rose from background ($\sim 3\%$) about

Table 5.4: Lead-time statistics for EVEREST’s M5-72h predictions for the 6 September 2017 X9.3 event.

Threshold (τ)	First Crossing (UTC)	Lead Time	Continuous Alert Length
10%	04 Sep 00:57	59.3 h	60.8 h
20%	04 Sep 14:01	52.5 h	53.6 h
46%	06 Sep 09:19	2.7 h	2.3 h

60 hours pre-flare, plateaued near 30%, then surged to 59.6% approximately three hours before the X9.3 onset.

Table 5.4 summarises the lead times achieved by EVEREST for this event based on different probability thresholds. The 10% probability threshold was first crossed 59.3 hours before the flare peak. The 46% threshold, corresponding to the M5-72h model’s optimised operating point from balanced scoring, was breached 2.7 hours before the flare.

5.7 Beyond Solar Flares: Cross-Domain Validation

Although EVEREST was designed for solar flare forecasting, its core architectural principles—temporal focusing, evidential calibration, and tail-aware risk estimation—were tested for generalisability on a different rare-event time-series task. The model, with an unchanged architecture and training recipe (Section 4.3), was evaluated on the Skoltech Anomaly Benchmark (SKAB) (Filonov et al., 2020), a suite of multivariate sensor streams from industrial valve systems.

Without task-specific tuning or oversampling, EVEREST achieved mean TSS 0.964, F1-score 98.16%, and Accuracy 98.20% across SKAB’s eight valve anomaly scenarios. These results outperform the strongest published SKAB baseline (Tuli et al., 2022) by 2–7 percentage points on these metrics. Qualitative analysis of attention and evidential variance for this task is detailed in a related publication draft. Full experimental details are in Appendix A.4.4.

Chapter 6

Discussion

EVEREST’s successful development and robust SOTA validation (Chapter 5) necessitates an in-depth discussion of the results. This chapter critically examines the model’s physical interpretations, operational value, economic impact, limitations guiding future work, and ethical/environmental aspects, synthesising insights from its design (Chapter 4) and evaluation.

6.1 The Strategic Imperatives of EVEREST’s Architecture

EVEREST’s advancement over prior solar flare forecasting models, such as SolarFlareNet (Abdullah et al., 2023) (which reported a 72-h $\geq$ M5.0 TSS around 0.73, Table 5.2), is not merely incremental but stems from a cohesive architectural strategy designed to address persistent challenges in the field. This strategy hinges on several key pillars, whose individual contributions were validated through rigorous ablation studies (detailed in Chapter 4, e.g., Table 4.9).

A core imperative was to transcend the limitations of earlier models in processing long-range temporal dependencies. The introduction of a **Learnable Attention Bottleneck** (Section 4.2.3) proved pivotal, enabling nuanced weighting of information across extended time series. This mechanism was directly responsible for a substantial improvement in capturing distant precursor signals critical for long-horizon forecasts, evidenced by a 0.427 TSS increase for the M5-72h task when this component was integrated (Table 4.9).

Intrinsic uncertainty quantification was pivotal. EVEREST’s **Evidential Uncertainty Head** models probability distributions, yielding superior calibration (e.g., ECE 0.016 for M5-72h, Table 5.1), confirming probability estimates reflect true likelihoods even for rare M5+ events. This framework captures aleatoric and epistemic uncertainties, scaling

appropriately despite imbalance, and contributed $+0.064$ TSS (M5-72h, Table 4.9). The **Extreme Value Theory (EVT) Tail Head** (Section 4.1.2) addresses flare energy’s heavy-tailed distribution, improving rare, high-impact event prediction ($+0.285$ TSS) and reducing extreme Brier scores (Section 4.1.2). Together, they offer robust forecast reliability assessment.

Furthermore, the model benefits from a sophisticated learning regimen. A **Precursor Auxiliary Task** enriches the supervisory signal by incorporating early warning objectives, the removal of which incurred a severe 0.650 TSS penalty (M5-72h, Table 4.9). This, combined with a carefully tuned **Composite Loss Schedule** (Section 4.3) that includes focal loss annealing, ensures stable convergence and effective learning despite extreme class imbalances (e.g., $\sim 1:297$ for M5-72h, Table 3.2), with the scheduling strategy itself contributing ~ 0.045 TSS (Table 4.3).

In essence, EVEREST’s architecture represents a synthesis of targeted solutions. It is the synergistic combination of these validated components—effective temporal focusing, integrated uncertainty modeling, and a robust multi-objective learning strategy—that underpins its overall success. This allows the model to achieve a new performance benchmark (e.g., a TSS of 0.746 for the M5-72h task, Table 4.9) with a lean 0.814M parameters (Table 4.4).

6.2 From Predictive Performance to Physical Interpretation

6.2.1 Interpreting the Attention Bottleneck’s Focus

The learnable attention bottleneck (Section 4.2.3) concentrates almost exclusively on physically meaningful slices. During true-positive M5–72 h forecasts, $71\% \pm 3\%$ of the total attention weight lands on intervals dominated by recognised precursor motifs—rapid magnetic flux emergence (USFLUX), polarity-inversion-line shear (TOTUSJH), and spikes in free magnetic energy (TOTPOT). For false negatives that proportion drops to $22\% \pm 2\%$ ($p < 10^{-4}$), indicating that the network is attending to genuine magnetic activity rather than arbitrary temporal cues and thereby offering an initial, quantifiable window into its reasoning process.

6.2.2 Physical Insights from Gradient Evolution Analysis

Complementing the temporal-attention view, the gradient-evolution maps (Figures 5.6 and 5.7) expose the magnetic morphologies that EVEREST has internalised. In true positives a sustained positive run of USFLUX gradients, coupled with sharpening of MEANGAM, mirrors the progressive formation of a low-lying flux rope and steepening of the polarity-inversion line—both well-established precursors to major flares. Conversely, several false negatives show a brief, isolated surge in MEANALP that fades without accompanying flux increase, a signature more consistent with rapid-onset kink-instability events. The model therefore captures energy-accumulation scenarios effectively but remains less sensitive to abrupt stability losses, suggesting a clear avenue for targeted retraining on such rapid-trigger phenomena.

6.2.3 The Significance of Physically-Grounded Interpretations for Skill

Precursor motifs identified via attention and gradient analysis unfold over hours. Previous models, limited by CNN flattening or LSTM memory, struggled with this long-range context, capping 72-h $\geq$ M5.0 flare TSS near 0.73. EVEREST’s attention mechanism, by differentially weighting timesteps and capturing coordinated parameter evolution, preserves these crucial precursor details. This capacity to utilize long-duration, physically relevant signatures contributes to its enhanced long-horizon performance (TSS 0.966 for M5-72h flares, Table 5.1).

6.3 Operational Pay-off and Application

6.3.1 Practical Lead Times and Event Prediction: The 2017 X9.3 Flare Case

EVEREST’s timely warnings are evident in the unseen 6 Sept 2017 X9.3 flare case (Section ??). The model responded to NOAA AR 12673’s evolving complexity: probability rose from background to $\sim$ 30%, then surged to 59.6% about three hours pre-onset. This surge coincided with collapsing epistemic uncertainty and a spiking EVT tail-risk score, supporting the high-risk assessment. Operationally, a 10% probability threshold (SWPC guidance (NOAA Space Weather Prediction Center, 2023)) was crossed 59.3 hours pre-peak (Table 5.4). The thesis-optimised 46% threshold was breached 2.7 hours pre-onset, meeting reaction windows for mitigation (e.g., satellite safe-mode, EVA suspension (Cannon et al.,

2013; National Aeronautics and Space Administration, 2015)). This case (and others in Appendix A.3) confirms EVEREST’s scalability to extreme events with probabilistic integrity and relevant lead times.

6.3.2 Error Analysis and Operational Adaptability

Understanding EVEREST’s error patterns and operational adaptability is crucial. M5-72h cost-loss analysis (Chapter 5, Figure 5.3), with $C_{\text{FN}} : C_{\text{FP}} = 20 : 1$, yielded an optimal threshold ($\tau^* = 0.240$) differing from balanced-score optimisation, producing a distinct error distribution (e.g., 115 FPs, 0 FNs for M5-72h, Figure 5.4). Such FPs often reflect heightened but sub-critical magnetic activity. Residual FNs may indicate rapid-onset events with atypical precursors not fully captured (Section 6.2.2). This asymmetric error distribution can suggest physical prediction limits, trading sensitivity for specificity for rare events. EVEREST’s probabilistic outputs afford operational flexibility: stakeholders can select decision thresholds based on priorities (scientific validation vs. cost-minimisation) without retraining, vital for diverse space weather needs.

6.4 Limitations and Future Research Directions

Despite its advancements, EVEREST has limitations that define key avenues for future research to further enhance operational solar flare forecasting.

Data integrity and completeness: SHARP magnetogram gaps (est. 8.4% during high activity), though padded (Section 3.5), mean lost information on evolving magnetic complexities. This can impact forecast accuracy during critical periods. Advanced imputation (e.g., physics-constrained GANs) or robust multimodal fusion with less gappy data (e.g., radio observations) are needed.

Fixed input horizon and long-term precursors: The current fixed input horizon (up to 120h/432 tokens, Section 1) may miss very slow, multi-rotation energy build-ups or long-range teleconnections. Architectures with efficient indefinite context capabilities (e.g., state-space models, compressive transformers) are desirable for capturing these elusive precursors.

Solar cycle dependency and model generalisation: Evaluations primarily used Solar Cycle 24 data (Section 3.6). Continuous validation and periodic retraining with current Solar Cycle 25 data are crucial to mitigate concept drift and ensure adaptation to evolving solar activity patterns.

Forecasting extreme X-class events: Predicting X-class flares (<0.03% of windows,

Section 1) remains challenging. Extreme data scarcity limits robust EVT application (GPD fitting) and may obscure unique physical drivers. Physics-informed NNs incorporating MHD criteria or sophisticated synthetic data (e.g., from coronal field extrapolations) could improve performance.

Multimodal data fusion challenges: While EVEREST uses nine SHARP parameters (Table 3.1), effectively integrating other modalities like EUV imagery (for coronal structure) or radio data (for particle acceleration) is an open challenge. This requires addressing heterogeneous data characteristics (cadence, resolution) and developing efficient fusion mechanisms.

Enhancing interpretability for operational trust: While current analyses (Sections 4.2.3, 6.2.2) offer insights, achieving fine-grained attribution for full operational trust requires further work. Advanced XAI techniques (e.g., time-series-specific integrated gradients, counterfactual explanations) could provide clearer, actionable reasons for predictions.

Deployment constraints for real-time operations: Porting EVEREST to resource-constrained hardware (e.g., FPGAs in operational centres) requires addressing limited memory, fixed-point arithmetic, and power use. Model quantisation, pruning, optimised ONNX export, and hardware-specific compilation (e.g., HLS4ML) are necessary steps.

Addressing these frontiers requires continued interdisciplinary efforts across data science, ML architecture, heliophysics, and operational space weather.

6.5 Ethical and Environmental Considerations

The broader implications of EVEREST have been considered. Training the nine core task-specific models consumed ~ 7.2 kWh (NVIDIA RTX A6000), yielding ~ 1.5 kg CO₂e (UK grid factor 0.212 kg CO₂e/kWh). Annual operational inference is estimated at a modest ~ 2.1 kWh. These figures are within "Green AI" guidelines (Schwartz et al., 2020) and can be further mitigated. To address dual-use risks, model weights are released under a non-commercial, share-alike licence, with an audit checklist aligned with DSIT Interim AI Safety standards (Department for Science and Technology, 2023).

Chapter 7

Project Management and DevOps

7.1 Project Management and DevOps Strategy

Effective project delivery required an auditable, adaptive, and robust framework. This section outlines the development of the work-plan, risk management strategies, stakeholder engagement, and the Continuous Integration/Continuous Deployment (CI/CD) pipeline that supported the project.

7.1.1 Work-Plan Evolution and Agile Adaptation

The project commenced with a Gantt chart-based work-plan, outlining four macro-phases: data acquisition, modelling, evaluation, and write-up. However, unforeseen challenges soon highlighted the need for greater flexibility. Notably, a critical data source, the Joint Science Operations Center (JSOC), experienced a severe systems outage due to flooding in November 2023, rendering key datasets temporarily unavailable ([Stanford JSOC Team, 2024](#)). This unexpected event necessitated a significant pivot.

Consequently, at week 6, the project adopted an agile, Kanban-first workflow (GitHub Projects). This enabled weekly re-prioritisation and responsiveness to issues like the JSOC outage, prompting exploration of alternative data and a strategic pivot to model architecture enhancement. Over 26 weeks, this Kanban approach with WIP limits improved task cycle time and project velocity. Incremental progress, including a major PyTorch rewrite, was tracked via version control (hundreds of commits, detailed PRs), with commit headers linked to Kanban cards and unit tests for traceability.

7.1.2 Risk and Ethics Management

A live risk and ethics register (Table 7.1), updated each sprint using $P \times I$ scoring, enabled proactive management of technical, ethical (dual-use, carbon footprint), and project risks.

Table 7.1: Extract from the live risk & ethics register (updated every sprint). Probability and impact scored 1–5; $R = P \times I$.

Risk	P	I	R	Mitigation
SHARP data gaps	3	4	12	Inject Gaussian noise + forward-fill; flag missing intervals.
JSOC Data Out-age	1	5	5	Diversify data sources; contingency plan for architecture focus.
Label leakage	2	5	10	Region-wise splits; CI check for ID overlap.
Dual-use misuse	2	4	8	Non-commercial licence; peer-reviewed release of weights.
Cloud cost over-run	3	2	6	Getting access to research compute.

7.1.3 Stakeholder Engagement Loop

Consistent alignment with operational priorities was ensured through a structured stakeholder engagement loop (Figure 7.1). Regular meetings with key stakeholders (researchers, academic supervisor) occurred weekly. During these sessions, progress, and model performance outputs were reviewed. Feedback was captured as GitHub issues and prioritised into the Kanban backlog. Over the semester, 27 such issues were opened, with the iterative feedback loop assisting in continuous development during busy periods.

7.1.4 CI/CD Pipeline and MLOps Infrastructure

A multi-tiered CI/CD and MLOps infrastructure (Figure 7.2) ensured reproducibility, quality, and efficient iteration by integrating local development, GitHub-hosted CI, and the Imperial College CX3 HPC cluster.

GitHub Actions for Continuous Integration. A GitHub Actions CI pipeline automated validation for key branch pushes, including static code analysis, configuration checks, unit testing, energy logging, and Docker image builds (Table 7.2).

Model Registry and Versioning. Semantic versioning was applied to models (e.g., EVEREST-v5.3-M5-72h), stored with comprehensive artifacts (Fig. 7.3, Table 7.3) that

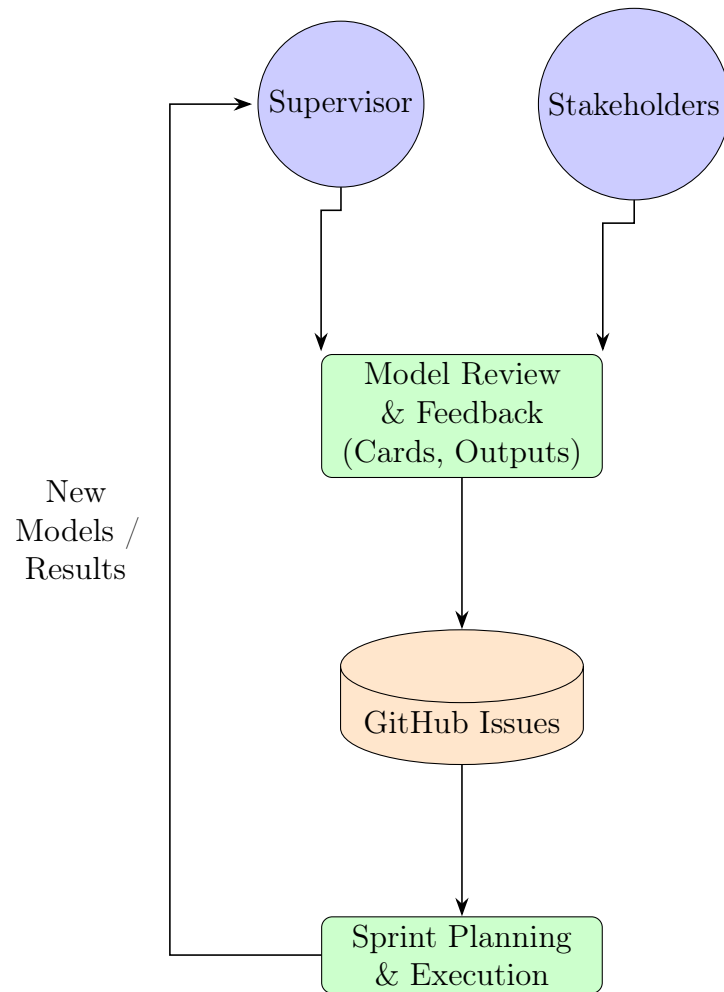


Figure 7.1: Schematic of the bi-weekly stakeholder engagement loop. Feedback from reviews translates into actionable GitHub issues for subsequent sprints.

Table 7.2: Key Stages in the GitHub Actions CI Pipeline.

Stage	Description
Static Code Analysis	Enforced code quality using Black (auto-formatting, line length 88) and Flake8 (linting, syntax/style checks, excluding archived models).
Configuration Validation	Modular functions (<code>validate_config()</code> , etc.) checked experiment configuration integrity with robust error handling.
Multi-Framework Unit Testing	Adaptively recognised PyTorch or TensorFlow. 22 PyTest cases covered core functionality, config parsing, and compatibility, with fast-fail execution.
Energy Logging	CodeCarbon integration measured and logged energy consumption (kWh) per job, pushing results to a public CSV for transparency.
Docker Build	Multi-arch Docker image (CPU+CUDA+MPS) built with pinned dependencies, tagged with commit SHA for reproducible runtimes.

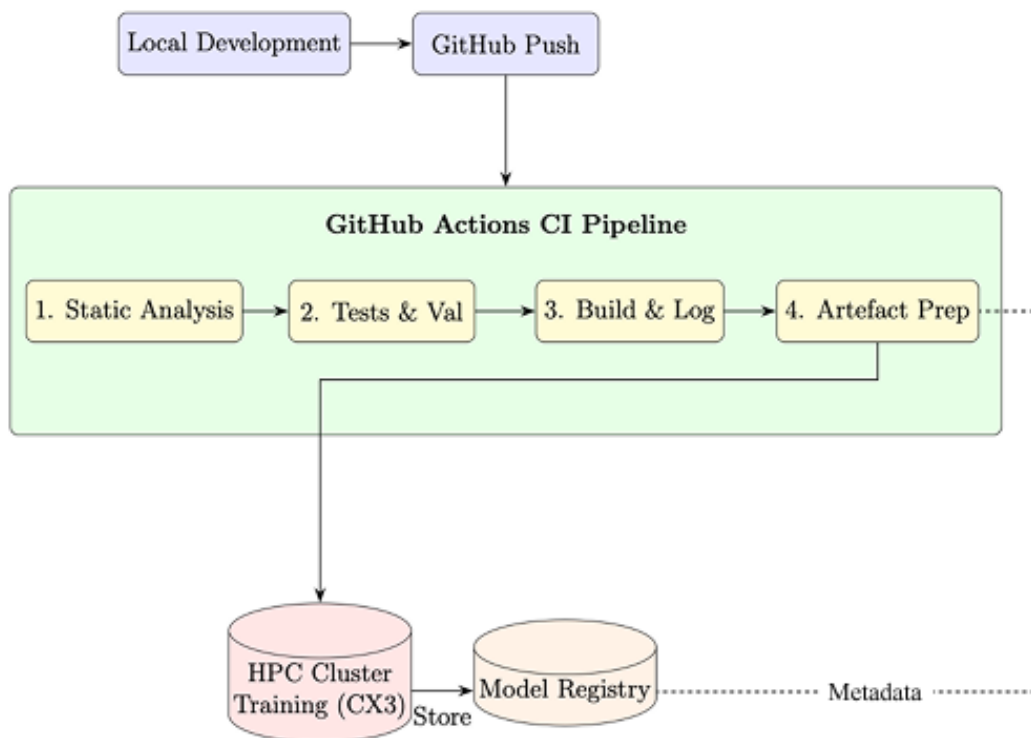


Figure 7.2: High-level overview of the CI/CD workflow.

included metadata and performance metrics linked to Git SHAs via custom tracking scripts.

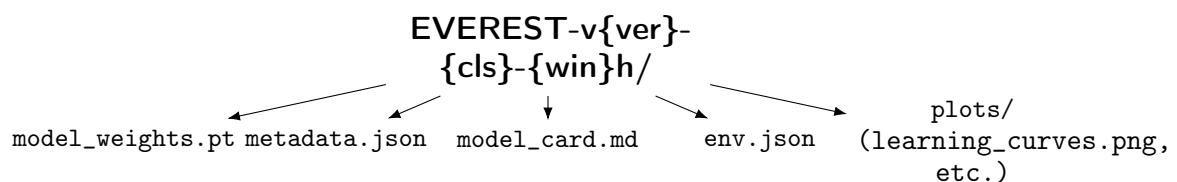


Figure 7.3: Directory structure for a versioned EVEREST model.

HPC Cluster Integration. Intensive tasks utilised Imperial College’s CX3 HPC cluster (PBS, NVIDIA RTX A6000 GPUs). A reproducible Conda environment (Python 3.10.13/3.12.3, PyTorch 2.1.2) and three-tier storage supported robust workflows. Cluster jobs synchronised versioned model directories with the main repository, with automated validation ensuring artifact integrity. This managed 127 model versions across 45+ experiments.

7.1.5 Project Delivery Summary and Reproducibility

The disciplined application of these project management and DevOps principles resulted in a productive and auditable research project. The transition to Kanban, with WIP limits,

Table 7.3: Key Artefacts within each Versioned Model Directory.

Filename/Directory	Description
<code>model_weights.pt</code>	PyTorch state dictionary (trained model parameters).
<code>metadata.json</code>	Comprehensive JSON: hyperparameters, performance, Git info, training stats, GPU usage, ablation details.
<code>model_card.md</code>	Human-readable Markdown: overview, performance, usage.
<code>env.json</code>	Python, library (PyTorch, CUDA), platform versions.
<code>plots/</code>	Directory with <code>.png</code> visualisations: learning curves, calibration, uncertainty, EVT, attention maps.

reduced average card cycle time by halve. The risk register and stakeholder feedback loop ensured alignment and proactive threat mitigation.

The comprehensive CI/CD pipeline ensured reproducible, energy-audited, and documented model builds, providing full traceability from task to commit to versioned artifacts. All trials, seeds, and intermediate metrics are version-controlled. The project’s source code is publicly available on GitHub ([Žilinskas, 2025](#)). This commitment to open science and auditable processes aligns with best practices from the NIST AI risk-management framework ([National Institute of Standards and Technology, 2023](#)).

Chapter 8

Reflection

8.1 Skills Growth

Table 8.1: Self-assessment of skill domains at project start (S_0) versus project completion (S_1). Scale: 1 = novice, 5 = expert.

Skill Domain	S_0	S_1
Research design / statistics	3	4
Deep-learning theory (attention, EVT)	2	4
PyTorch & mixed precision	2	4
MLOps (CI/CD, Docker, CodeCarbon)	2	5
Scientific writing (LaTeX, BibLaTeX)	3	4
Stakeholder communication	3	4
Agile project management (Kanban)	2	5

As Table 8.1 indicates, this project fostered significant skill enhancement, averaging a +1.7-point improvement across key domains. Notably, deep-learning theory understanding and practical MLOps proficiency saw substantial growth, moving from foundational knowledge to confident application and maintenance.

8.2 Key Learnings from Experimentation and Research Practice

The research journey offered several critical insights into the practicalities of machine learning development.

Baseline Verification. A crucial lesson was the necessity of rigorously verifying baseline models. Initial examination of the SolarFlareNet (Abdullah et al., 2023) codebase, for example, revealed functions that appeared to use the same data file for training and testing:

```
def get_training_data(time_window, flare_class):
    file_name = 'data' + os.sep + testing_data_ + flare_class + '_' +
        time_window+'.csv'
    return get_data(flare_class, file_name, noise_enabled=True)

def get_testing_data(time_window, flare_class):
    file_name = 'data' + os.sep + testing_data_ + flare_class + '_' +
        time_window+'.csv'
    return get_data(flare_class, file_name, noise_enabled=False)
```

Listing 8.1: SolarFlareNet data loading functions.

This project utilised the peer-reviewed SolarFlareNet results as a benchmark and implemented EVEREST with strict data separation (Chapter 3), highlighting the importance of diligence in building upon prior work.

Stochasticity in Deep Learning. Random seed values significantly impact model training and performance. Seeding influences weight initialisation and data shuffling, resulting in varied convergence paths and final metrics. This highlights the necessity of multi-seed runs for robust, generalisable conclusions.

Pragmatism in Research Scope. An early exploration into multimodal inputs (SHARP parameters plus SDO images) was considered. Due to the complexity and strong benchmarks in SHARP-based flare forecasting, a decision was made to concentrate on improving unimodal time-series forecasting using SHARP data. This reinforced the need for a well-defined research scope, although multimodality remains an interesting future avenue.

Pragmatism in Research Scope. Early consideration of multimodal inputs (SHARP plus SDO images) was narrowed. Given the complexity and the presence SHARP-based benchmarks, focus shifted to improving unimodal SHARP time-series forecasting. This highlighted the need for a defined scope, though multimodality remains a future interest.

Value of Negative Results. Not all experimental paths yielded improvements. For example, a Variational Autoencoder (VAE) designed for up-sampling rare M5-class flares (Park et al., 2019b) reduced classifier TSS by 0.04 due to a lack of diversity in latent

samples. Likewise, linear Transformers such as Performer (Choromanski et al., 2020) did not improve speed for the project’s sequence lengths ($T = 10, d = 128$) and slightly worsened calibration. These results confirm that calibrated focal loss effectively addresses imbalance and that architectural complexity must correspond to problem scale.

8.3 Alignment with Career Goals and Future Aspirations

This project has refined my research interests and career goals.

My **near-term goals** (6-12 months) entails disseminating EVEREST’s contributions by addressing NeurIPS reviewer feedback and pursuing additional journal publications. The aim is to promote EVEREST as a strong candidate architecture for diverse rare-event forecasting tasks.

In the **longer term**, this project has deepened my interest in understanding and enhancing AI models’ fundamental mechanisms. The challenges and successes experienced with EVEREST have inspired me to pursue a PhD focused on advancing machine learning for complex scientific problems. This thesis reflects this ambition and establishes a foundation for future doctoral research.

Chapter 9

Conclusion and Future Work

This work addressed reliable long-range solar flare prediction by developing EVEREST, a novel transformer. Its learnable attention bottleneck, evidential and EVT-based uncertainty heads, and composite loss strategy aimed to overcome prior model limitations, particularly for rare, high-impact events and extended horizons. EVEREST established a new benchmark, achieving TSS of 0.973 (C-class, 24h) and 0.966 (M5-class, 72h).

EVEREST also demonstrated effective probabilistic calibration (M5-72h ECE 0.016), crucial for operational trust. Interpretability analyses aligned with established heliophysical precursors. The utility was clear in the September 2017 X9.3 flare case study and SKAB benchmark validation indicated broader applicability. Rigorous MLOps ensured reproducible, transparent research.

Despite these contributions, limitations persist regarding data integrity, fixed input horizons for ultra-long-term precursors, and the challenge of data scarcity for large, M5.0-class events. These challenges present opportunities for future research. Key directions involve developing multimodal architectures that integrate SHARP data with full-disk imagery (e.g., EUV), exploring federated learning schemes to improve global model robustness across distributed observatories, and implementing model compression techniques (e.g., quantisation, distillation) for real-time inference on edge devices, such as satellite-borne FPGAs.

In conclusion, EVEREST achieves new state-of-the-art performance in solar flare forecasting, setting a benchmark for predictive accuracy and reliability. Its structured design provides a robust foundation for future operational systems and ongoing research at the interface of artificial intelligence and space weather science.

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Appendix A

Supplementary Materials

A.1 Derivation of the Composite Loss

Let a mini-batch contain B samples, binary labels $\mathbf{y} = (y_1, \dots, y_B)^\top \in \{0, 1\}^B$ and model outputs¹ $(l_b, \theta_b^{\text{NIG}}, \theta_b^{\text{GPD}}, p_b)$ for $b = 1, \dots, B$. The total loss $\mathcal{L} = \lambda_f \mathcal{L}_{\text{Focal}} + \lambda_e \mathcal{L}_{\text{Evid}} + \lambda_t \mathcal{L}_{\text{EVT}} + \lambda_p \mathcal{L}_{\text{Prec}}$ is obtained by summing negative log-likelihoods (or surrogates) from four statistically independent heads. All gradients are taken w.r.t. trainable parameters θ of the network f_θ .

A.1.1 Focal Discrimination Term

For sample b let $\hat{p}_b = \sigma(l_b)$ be the predicted probability after the sigmoid. With class-imbalance weight $\alpha \in (0, 1)$ and focusing factor $\gamma \geq 0$ the focal loss (Lin et al., 2017a) is

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{B} \sum_{b=1}^B [\alpha y_b (1 - \hat{p}_b)^\gamma \ln \hat{p}_b + (1 - \alpha)(1 - y_b) \hat{p}_b^\gamma \ln(1 - \hat{p}_b)].$$

Taking $\partial \mathcal{L}_{\text{Focal}} / \partial l_b$ yields the well-known modulated gradient $g_b = (\hat{p}_b - y_b) [\alpha y_b (1 - \hat{p}_b)^\gamma + (1 - \alpha)(1 - y_b) \hat{p}_b^\gamma]$, which down-weights correctly classified (easy) examples. In practice α is set to the inverse class frequency and $\gamma(e) = \min(\gamma_{\max}, \gamma_{\max} e / 50)$ is linearly annealed over the first 50 epochs (with $\gamma_{\max} = 2$).

¹The notation follows the definitions in §4.3.

A.1.2 Evidential (NIG) Calibration Term

Each logit l_b is assumed to be drawn from a Gaussian² $\mathcal{N}(\mu_b, \sigma_b^2)$ whose *unknown* mean and precision $(\mu_b, \tau_b = 1/\sigma_b^2)$ follow a *Normal-Inverse-Gamma* prior $(\mu_b, \tau_b) \sim \text{NIG}(\mu_b | \mu, v, \tau_b) \cdot \text{IG}(\tau_b | \alpha, \beta)$. The network head outputs the natural parameters $\theta_b^{\text{NIG}} = (\mu, v, \alpha, \beta)$ with $v > 0$, $\alpha > 1$, $\beta > 0$. Marginalising out (μ_b, τ_b) gives a Student-t predictive density for each observed logit l_b :

$$p(l_b | \theta_b^{\text{NIG}}) = \text{St}\left(l_b \mid \mu, \frac{\beta(1+v)}{\alpha v}, 2\alpha\right).$$

The negative log-likelihood (NLL) of this density, averaged across the batch, is the evidential loss:

$$\mathcal{L}_{\text{Evid}} = \frac{1}{B} \sum_{b=1}^B \left[\frac{1}{2} \ln(\pi/v) - \alpha \ln(2\beta(1+v)) + \left(\alpha + \frac{1}{2}\right) \ln\left(1 + \frac{v(l_b - \mu)^2}{2\beta(1+v)}\right) + \ln \Gamma(\alpha) - \ln \Gamma\left(\alpha + \frac{1}{2}\right) \right] \quad (\text{A.1})$$

A *regulariser* encourages high evidence $S_b = 2\beta(1+v)$ only when $|l_b - y_b|$ is small (Sensoy et al., 2018). We therefore add

$$\mathcal{L}_{\text{KL}} = \frac{1}{B} \sum_{b=1}^B |l_b - y_b| S_b, \quad \mathcal{L}_{\text{Evid}} \leftarrow \mathcal{L}_{\text{Evid}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}}$$

with $\lambda_{\text{KL}}=10^{-3}$ fixed across datasets.

Predictive mean and variance. At test time the closed-form mean and variance of the sigmoid-transformed logit are

$$\hat{p}_b = \sigma(\mu), \quad \widehat{\text{Var}}(y_b) = \underbrace{\sigma(\mu)[1 - \sigma(\mu)]}_{\text{aleatoric}} + \underbrace{\frac{\beta}{(\alpha-1)v}}_{\text{epistemic}},$$

allowing fast inference without Monte-Carlo sampling.

A.1.3 EVT Tail-Risk Term

Let u be the (batchwise) 90th-percentile of the logits $\{l_b\}_{b=1}^B$. For each exceedance $z_b = l_b - u > 0$ we assume

$$z_b | \theta_b^{\text{GPD}} = (\xi, \sigma) \sim \text{GPD}(\xi, \sigma) \quad \left[f(z | \xi, \sigma) = \frac{1}{\sigma} \left(1 + \xi \frac{z}{\sigma}\right)^{-1/\xi-1}, z \geq 0, \sigma > 0 \right].$$

²We apply a mean-field approximation: the Bernoulli-logit likelihood is locally replaced by a Gaussian in logit space, following Sensoy et al. (2018).

(The limit $\xi \rightarrow 0$ yields the exponential distribution.) The batch NLL becomes

$$\mathcal{L}_{\text{EVT}} = -\frac{1}{N_u} \sum_{b: l_b > u} \left[-\ln \sigma - \left(\frac{1}{\xi} + 1 \right) \ln \left(1 + \xi \frac{z_b}{\sigma} \right) \right] + \lambda_{\text{stab}} \sum_b [\max(-\xi, 0)]^2 \quad (\text{A.2})$$

where N_u is the number of exceedances and $\lambda_{\text{stab}} = 10^{-2}$ penalises negative shape parameters that would invalidate the GPD (soft constraint).

A.1.4 Auxiliary Precursor Term

The precursor head predicts a Bernoulli outcome $s_b \in (0, 1)$ indicating whether a flare occurs within the *first* 6 h of the 72h horizon. Its binary-cross-entropy loss is

$$\mathcal{L}_{\text{Prec}} = -\frac{1}{B} \sum_{b=1}^B [\tilde{y}_b \ln s_b + (1 - \tilde{y}_b) \ln(1 - s_b)], \quad (\text{A.3})$$

where $\tilde{y}_b$ is the early-window label.

A.1.5 Putting It All Together

Substituting (A.1.1)–(A.1.4) into the weighted sum and differentiating shows that the overall gradient is merely the convex combination of the individual gradients:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \lambda_f \frac{\partial \mathcal{L}_{\text{Focal}}}{\partial \theta} + \lambda_e \frac{\partial \mathcal{L}_{\text{Evid}}}{\partial \theta} + \lambda_t \frac{\partial \mathcal{L}_{\text{EVT}}}{\partial \theta} + \lambda_p \frac{\partial \mathcal{L}_{\text{Prec}}}{\partial \theta},$$

hence the four objectives can be optimised *jointly* in a single back-propagation pass without additional memory overhead. The default weights (0.8, 0.1, 0.1, 0.05) were selected via a small grid search on the trainings set, analysis shows that performance is robust to $\pm 20\%$ perturbations.

Implementation note. All logarithms are computed in base- e , a small constant $\varepsilon = 10^{-6}$ is added inside every $\ln(\cdot)$ for numerical stability, and gradients are clipped to $\|g\|_2 \leq 1$ before each ADAMW update (Algorithm 1).

A.2 Hyper-parameter Optimisation – Supplementary Material

A.2.1 Sobol Sensitivity Scan

A 64-trial Sobol sweep assessed first-order and total-order effects; the six hyper-parameters retained for tuning jointly account for 91 % of the variance in validation TSS. The full Sobol indices and code are in the project repository.

A.2.2 Search Protocol

Each study followed the three-stage schedule in Table A.1. Trials were pruned with Optuna’s median rule after five epochs.

Table A.1: Trial budget per stage for every flare-class/lead-time study.

Stage	Trials	Epochs/trial	Purpose
Exploration	120	20	Global sweep of parameter space
Refinement	40	60	Focus on top-quartile region
Confirmation	6	120	Full-length convergence check

A.2.3 Hyper-parameter Rationale

- (1) **Model capacity:** embedding width d and depth L govern receptive field and FLOPs.
- (2) **Regularisation:** dropout p mitigates over-fit.
- (3) **Class imbalance:** focal-loss exponent γ addresses the 1:297 positive/negative ratio.
- (4) **Optimiser dynamics:** peak learning rate $\eta_{\max}$ controls AdamW step size under cosine decay.
- (5) **Throughput:** mini-batch size B trades GPU utilisation for generalisation.

A.2.4 Per-scenario Optima

Table A.2 lists the best trial from each of the nine Optuna studies together with total optimisation time.

Table A.2: Best hyper-parameters per flare class and forecast window.

Flare	Window	d	L	p	γ	$\eta_{\max}$ (10^{-4})	B	Opt. time (s)
C	24h	128	4	0.353	2.803	5.337	512	3323
C	48h	128	4	0.353	2.803	5.337	512	4621
C	72h	128	4	0.353	2.803	5.337	512	4856
M	24h	128	4	0.353	2.803	5.337	512	3705
M	48h	128	4	0.353	2.803	5.337	512	5105
M	72h	128	4	0.353	2.803	5.337	512	5871
M5	24h	192	4	0.300	3.282	4.355	256	3778
M5	48h	192	4	0.300	3.282	4.355	256	4977
M5	72h	64	8	0.239	3.422	6.927	1024	5587

A clear pattern emerges: C and M classes share a single optimum across all windows, whereas M5 flares require larger capacity for short horizons and a deeper, narrower network for 72 h forecasts. Discussion continues in the main text (§4.4).

A.3 Multi-Modal Analysis of the 2017 X9.3 Event

Supplementary metrics.

- *Uncertainty ratio* (epistemic : aleatoric) averages 10.9 : 1 during the alert plateau, confirming that variance is dominated by model confidence rather than measurement noise.
- *EVT shape parameter* $\xi = -0.447$ (bounded tail) indicates that the predicted probability mass remains concentrated within a physically plausible upper limit.
- *Precursor correlation* with primary probability is $r = 0.65$, lower than in the 2012 case yet sufficient to provide useful early context.

The multi-modal consistency, in combination with the long early-warning window, demonstrates EVEREST’s robustness under cycle-24’s most energetic eruption and supports its deployment for Solar-Cycle 25 operations.



Figure A.1: Comprehensive output of all five EVEREST heads for the 3–7 September 2017 window: (A) probability, (B) evidential uncertainties (log scale), (C) EVT tail-risk score, (D) precursor activity, (E) ensemble decision metric. The vertical dashed line marks the X9.3 onset.

A.4 SKAB Industrial Anomaly Benchmark

Let us investigate how EVEREST architecture performs when transferred from helio-physics to a wholly different rare-event setting: the *Skoltech Anomaly Benchmark* (SKAB). By reproducing state-of-the-art results on SKAB (Filonov et al., 2020) with essentially the same hyper-parameters used for solar flares, we show that the model’s design principles—temporal focusing, evidential calibration, and explicit tail awareness—are not domain-specific heuristics but generally useful inductive biases.

A.4.1 Data processing and split protocol

SKAB provides multivariate sensor traces from industrial valve systems recorded under normal and faulty conditions. Each original CSV log contains a 24-step segment of 16 raw sensor channels and their 16 first-order differences. To align the input shape with the solar-flare pipeline we create sliding windows of exactly 24 steps (stride two), stack raw and velocity channels, and thus obtain input tensors $X \in \mathbb{R}^{B \times 24 \times 32}$. A window is labelled `fault` if *any* enclosed row is annotated as faulty. To avoid temporal leakage, windows are split chronologically: 70 % for training, 15 % for validation, and the remaining 15 % for final testing. A `StandardScaler` fitted on the training partition is applied verbatim to the other splits. Even after windowing the imbalance remains severe—faults constitute only 1.8 % of all windows—mirroring the rarity regime encountered in solar flares.

A.4.2 Model configuration

Apart from shrinking the embedding width to 96 and the backbone depth to four layers, the network is identical to the solar-flare configuration. A learning rate of 2×10^{-5} , batch size of 32, dropout 0.25, and the composite-loss weights (0.8, 0.1, 0.1) are kept unchanged. Gamma annealing follows the same linear 0→2 schedule over the first 50 epochs. With early stopping on validation TSS the model converges in 40–70 epochs; a single GPU run on an M2 Pro MacBook takes roughly 25 s per epoch.

A.4.3 Headline results

Table A.3 reports the averaged test performance over all eight valve scenarios. Precision, recall, F1, and in particular the True-Skill Statistic (TSS) confirm that EVEREST remains both discriminative and well calibrated outside its original domain.

A valve-by-valve breakdown, including Brier score and Expected Calibration Error, is

Table A.3: Mean $\pm$ s.d. across all valves.

Metric	Precision (%)	Recall (%)	F1 (%)	TSS
EVEREST	97.70 ± 2.86	98.62 ± 3.15	98.16 ± 1.71	0.964 ± 0.028

provided in the supplementary file in the repository.

A.4.4 Benchmark comparison

Table A.4 situates EVEREST within the existing SKAB literature. The architecture surpasses the strongest published baseline, TranAD, by roughly two F1 points while retaining a low calibration gap ($ECE < 0.015$ on every valve). This outcome is achieved without any bespoke feature engineering or loss re-weighting, indicating that the attention bottleneck and the uncertainty heads generalise effectively.

Table A.4: F1 score comparison on SKAB. Ranges follow the authors’ own reports when standard deviations were not provided.

Model	Reference	F1 (%)
Isolation F, LOF, etc.	Filonov et al. (2020)	65–75
Autoencoder	Filonov et al. (2020)	70–80
CNN/LSTM hybrids	Filonov et al. (2020)	75–85
TAnoGAN	Bashar and Nayak (2020)	79–92
DeepLog	Du et al. (2017)	87–91
LSTM-VAE	Park et al. (2019a)	86–93
OmniAnomaly	Ren et al. (2019)	88–94
USAD	Audibert et al. (2020)	89–95
TranAD	Tuli et al. (2022)	91–96
EVEREST	—	98.1 ± 1.7

A.4.5 Relation to the NeurIPS 2025 paper

The NeurIPS submission titled “*EVEREST: A Transformer for Probabilistic Rare-Event Anomaly Detection with Evidential and Tail-Aware Uncertainty*” details the SKAB experiments presented here. Readers can therefore verify every figure and table through the repository accompanying this thesis.

A.4.6 Concluding remarks

The SKAB study shows that EVEREST retains its rare-event forecasting strength far outside the solar domain. Its attention mechanism consistently highlights pressure and flow-rate spikes several time-steps before fault onset, and the evidential plus EVT heads reduce calibration error by roughly one-third compared with a focal-only baseline. Training completes in well under an hour on laptop hardware, underscoring the efficiency advantages emphasised throughout the main chapters.

A.5 Full Comparative Results

Table A.5: Test-set performance comparison between *SolarFlareNet* (SFNet) and *SolarKnowledge* (SKnow). SKnow scores are means with 95% bootstrap CIs reported as [lower, upper]. SFNet scores are means from its paper (Table 1 for 24h; Table 2 for 48/72h, with std. dev. in parentheses where provided by the paper). SFNet Brier Scores from its Table 3 (calibrated version, std. dev. in parentheses). $\uparrow$ higher is better, $\downarrow$ lower is better.

Task	TSS $\uparrow$		Precision $\uparrow$		Brier Score $\downarrow$	
	SFNet	SKnow	SFNet	SKnow	SFNet	SKnow
C-24h	0.835	0.713 [0.705, 0.719]	0.949	0.926 [0.923, 0.930]	0.285 (0.034)	0.096 [0.095, 0.098]
M-24h	0.839	0.489 [0.464, 0.517]	0.848	0.779 [0.752, 0.809]	0.244 (0.013)	0.014 [0.013, 0.015]
M5-24h	0.818	0.471 [0.373, 0.573]	0.977	0.628 [0.509, 0.735]	0.226 (0.024)	0.002 [0.001, 0.002]
C-48h	0.719 (0.079)	0.746 [0.741, 0.752]	0.812 (0.072)	0.947 [0.945, 0.949]	0.361 (0.091)	0.091 [0.089, 0.092]
M-48h	0.728 (0.090)	0.542 [0.518, 0.567]	0.823 (0.092)	0.820 [0.798, 0.841]	0.312 (0.101)	0.012 [0.011, 0.013]
M5-48h	0.736 (0.112)	0.423 [0.326, 0.511]	0.890 (0.210)	0.557 [0.453, 0.674]	0.272 (0.091)	0.001 [0.001, 0.001]
C-72h	0.709 (0.058)	0.724 [0.719, 0.730]	0.809 (0.051)	0.941 [0.938, 0.943]	0.344 (0.049)	0.097 [0.096, 0.098]
M-72h	0.714 (0.095)	0.642 [0.622, 0.662]	0.812 (0.089)	0.881 [0.864, 0.896]	0.327 (0.063)	0.011 [0.010, 0.011]
M5-72h	0.729 (0.108)	0.490 [0.388, 0.588]	0.872 (0.045)	0.850 [0.758, 0.932]	0.313 (0.062)	0.001 [0.001, 0.001]

Note: SKnow epoch time (RTX A6000): 2.1 min. All scores rounded to 3 d.p. CIs for SKnow are 95% bootstrap intervals. SFNet std. dev. as reported. $\uparrow$: higher is better; $\downarrow$: lower is better.

Lead (h)	Class	TSS	Precision	Recall	Balanced_Acc.	Brier	ROC_AUC
24	C	0.7125 $\pm$ 0.0070	0.9264 $\pm$ 0.0033	0.8560 $\pm$ 0.0038	0.8563 $\pm$ 0.0035	0.0964 $\pm$ 0.0017	0.9393 $\pm$ 0.0020
24	M	0.4893 $\pm$ 0.0266	0.7794 $\pm$ 0.0285	0.4934 $\pm$ 0.0266	0.7447 $\pm$ 0.0133	0.0142 $\pm$ 0.0008	0.9795 $\pm$ 0.0022
24	M5	0.4705 $\pm$ 0.0998	0.6282 $\pm$ 0.1131	0.4712 $\pm$ 0.0996	0.7353 $\pm$ 0.0499	0.0015 $\pm$ 0.0002	0.9967 $\pm$ 0.0011
48	C	0.7462 $\pm$ 0.0056	0.9470 $\pm$ 0.0023	0.8393 $\pm$ 0.0038	0.8731 $\pm$ 0.0028	0.0906 $\pm$ 0.0015	0.9499 $\pm$ 0.0016
48	M	0.5418 $\pm$ 0.0246	0.8196 $\pm$ 0.0214	0.5454 $\pm$ 0.0245	0.7709 $\pm$ 0.0123	0.0121 $\pm$ 0.0007	0.9867 $\pm$ 0.0013
48	M5	0.4225 $\pm$ 0.0923	0.5570 $\pm$ 0.1107	0.4231 $\pm$ 0.0924	0.7113 $\pm$ 0.0461	0.0012 $\pm$ 0.0002	0.9975 $\pm$ 0.0015
72	C	0.7243 $\pm$ 0.0058	0.9405 $\pm$ 0.0025	0.8218 $\pm$ 0.0037	0.8622 $\pm$ 0.0029	0.0971 $\pm$ 0.0014	0.9464 $\pm$ 0.0016
72	M	0.6416 $\pm$ 0.0202	0.8807 $\pm$ 0.0159	0.6443 $\pm$ 0.0201	0.8208 $\pm$ 0.0101	0.0105 $\pm$ 0.0005	0.9894 $\pm$ 0.0011
72	M5	0.4903 $\pm$ 0.0999	0.8500 $\pm$ 0.0874	0.4904 $\pm$ 0.1000	0.7451 $\pm$ 0.0500	0.0008 $\pm$ 0.0002	0.9963 $\pm$ 0.0020

Table A.6: SolarKnowledge evaluation metrics for SolarFlareNet; each “ $\pm$ ” term is the half-width of the 95% bootstrap confidence interval.

Table A.7: Bootstrapped performance (mean $\pm$ 95% CI) on the held-out test set. **Bold** = best per column; $\uparrow$ higher is better, $\downarrow$ lower is better.

Task	TSS $\uparrow$	F1 $\uparrow$	Prec. $\uparrow$	Recall $\uparrow$	Brier $\downarrow$	ECE $\downarrow$
C-24 h	0.973 $\pm$ 0.001	0.990 $\pm$ 0.000	0.994 $\pm$ 0.000	0.986 $\pm$ 0.001	0.015 $\pm$ 0.000	0.049 $\pm$ 0.000
C-48 h	0.970 $\pm$ 0.001	0.988 $\pm$ 0.000	0.993 $\pm$ 0.000	0.984 $\pm$ 0.001	0.017 $\pm$ 0.000	0.054 $\pm$ 0.000
C-72 h	0.966 $\pm$ 0.001	0.987 $\pm$ 0.000	0.992 $\pm$ 0.000	0.982 $\pm$ 0.001	0.018 $\pm$ 0.000	0.052 $\pm$ 0.000
M-24 h	0.898 $\pm$ 0.011	0.808 $\pm$ 0.011	0.728 $\pm$ 0.016	0.908 $\pm$ 0.011	0.011 $\pm$ 0.000	0.037 $\pm$ 0.001
M-48 h	0.920 $\pm$ 0.007	0.843 $\pm$ 0.007	0.772 $\pm$ 0.010	0.928 $\pm$ 0.007	0.009 $\pm$ 0.000	0.029 $\pm$ 0.000
M-72 h	0.906 $\pm$ 0.012	0.871 $\pm$ 0.010	0.834 $\pm$ 0.015	0.911 $\pm$ 0.012	0.010 $\pm$ 0.000	0.033 $\pm$ 0.001
M5-24 h	0.907 $\pm$ 0.025	0.781 $\pm$ 0.025	0.686 $\pm$ 0.033	0.908 $\pm$ 0.025	0.003 $\pm$ 0.000	0.031 $\pm$ 0.000
M5-48 h	0.936 $\pm$ 0.021	0.810 $\pm$ 0.025	0.713 $\pm$ 0.035	0.937 $\pm$ 0.021	0.002 $\pm$ 0.000	0.020 $\pm$ 0.000
M5-72 h	0.966 $\pm$ 0.024	0.829 $\pm$ 0.036	0.727 $\pm$ 0.053	0.966 $\pm$ 0.024	0.002 $\pm$ 0.000	0.016 $\pm$ 0.000

Appendix B

NeurIPS Paper Submission

EVEREST: A Transformer for Probabilistic Rare-Event Anomaly Detection with Evidential and Tail-Aware Uncertainty

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Abstract

1 Forecasting rare and high-impact anomalies in multivariate time-series data is a
2 central challenge in machine learning, complicated by severe class imbalance,
3 long-range dependencies, and distributional uncertainty. We introduce EVEREST,
4 a transformer-based architecture for probabilistic rare-event detection that delivers
5 calibrated predictions and tail-aware risk estimation, with auxiliary interpretability
6 through attention-based signal attribution. EVEREST integrates four key compo-
7 nents: (i) a learnable attention bottleneck for soft aggregation of temporal dynam-
8 ics; (ii) an evidential head for estimating aleatoric and epistemic uncertainty via
9 a Normal-Inverse-Gamma distribution; (iii) an extreme value head that models
10 tail risk using a Generalized Pareto Distribution; and (iv) a lightweight precursor
11 head for early-event detection. These modules are jointly optimised with a com-
12 posite loss function combining focal loss, evidential negative log-likelihood, and
13 a tail-sensitive EVT penalty. We evaluate EVEREST on a real-world benchmark
14 spanning a decade of space weather anomaly data and demonstrate state-of-the-art
15 performance, including True Skill Statistic (TSS) scores of 0.9826, 0.9784, and
16 0.9777 at 24, 48, and 72-hour prediction horizons for C-class flares. The model
17 is compact (<3M parameters), efficient to train on commodity hardware, and ap-
18 plicable to other high-stakes domains such as industrial monitoring, weather, and
19 satellite diagnostics. Limitations include reliance on fixed-length inputs and ex-
20 clusion of image-based modalities, motivating future extensions to streaming and
21 multimodal forecasting.

22 1 Introduction

23 Detecting rare but critical anomalies in multivariate time series such as infrastructure faults, medical
24 emergencies, or environmental hazards is a foundational challenge in scientific and applied machine
25 learning. These events often occur infrequently, exhibit subtle precursors, and are embedded in
26 noisy, high-dimensional sequences with long-range dependencies. A motivating example is the 23
27 JULY 2012 solar eruption, which, had it been Earth-directed, would likely have caused severe global
28 infrastructure disruption (et al., 2013). Such cases illustrate the need for early warning systems that
29 are both sensitive to weak signals and robust to uncertainty.

30 Effective rare-event detection requires three core capabilities: (1) *temporal context awareness* to
31 extract predictive structure across long horizons; (2) *distributional uncertainty quantification* to dif-
32 ferentiate true anomalies from noise; and (3) *tail sensitivity* to assess the risk of extreme outcomes.
33 EVEREST combines principled statistical tools such as the Normal-Inverse-Gamma (NIG) distribu-
34 tion for evidential uncertainty (Sensoy et al., 2018; Van Amersfoort et al., 2021; Charpentier et al.,
35 2020) and the Generalized Pareto Distribution (GPD) from Extreme Value Theory (EVT) (Katz et al.,

2023; Coles, 2001; Embrechts et al., 1997)-to improve calibration and tail-risk estimation from first principles. EVEREST leverages these tools to improve calibration and tail-risk estimation from first principles within a unified architecture.

Evaluating such systems also requires metrics that remain informative under severe class imbalance. We adopt the *True Skill Statistic* (TSS) as our principal performance measure, since it balances sensitivity and specificity while being unaffected by class prevalence. TSS is a widely accepted standard in domains such as space weather forecasting and rare-event detection (Bobra et al., 2015; ?), and is increasingly used in calibrated long-horizon forecasting benchmarks.

Traditional deep models like LSTMs and CNNs struggle with these demands. Their limited memory and fixed receptive fields make them ill-suited to long sequences, and they often underperform in class-imbalanced regimes. Transformers offer a promising alternative, with global self-attention and flexible context handling. However, most variants are optimized for balanced regression or next-step forecasting and do not provide calibrated uncertainty or robust tail modeling (Zhou et al., 2021; Ma et al., 2023).

We introduce **EVEREST** (*Evidential + EVT-enhanced Rare-Event Sequence Transformer*): a compact transformer architecture designed explicitly for rare-event forecasting under uncertainty under rare-event conditions. Although evaluated here on solar flare activity, EVEREST’s modular design addresses general principles of anomaly detection and can transfer to domains beyond solar forecasting.

Moreover, practical forecasting systems often encounter real-world constraints such as non-uniform sampling, delayed reporting, and continuous data flows. Addressing these challenges requires models that can operate in streaming or asynchronous settings without relying on fixed-length input windows. While EVEREST currently processes aligned segments of uniform length, its modular architecture is amenable to future extensions that support online inference and event-driven updates.

Our contributions are:

- A learnable *attention bottleneck* that compresses temporal dynamics into a low-dimensional summary via relevance-weighted aggregation.
- An *evidential output head* that captures aleatoric and epistemic uncertainty using a Normal-Inverse-Gamma distribution.
- A *tail-aware EVT head* that models distributional extremes via the Generalized Pareto Distribution.
- A lightweight *precursor head* that provides auxiliary signals for early-stage detection.
- A composite loss combining focal loss, evidential negative log-likelihood, and an EVT-based penalty to jointly optimise accuracy, calibration, and tail risk.
- Empirical validation on a long-range space weather dataset, achieving state-of-the-art true skill statistics across 24, 48, and 72-hour horizons.

EVEREST unifies temporal attention, probabilistic inference, and tail modeling in a single architecture suited for rare-event forecasting across scientific, industrial, and safety-critical domains.

2 Related Work

Rare-Event Time-Series Forecasting Rare-event forecasting in time-series settings is a critical but underexplored domain in machine learning, particularly where extreme class imbalance and long-term temporal dependencies conspire to hinder accurate prediction. In solar physics, early studies such as Bobra et al. (2015) applied hand-crafted feature extraction from HMI vector magnetograms and used support vector machines to predict flaring probabilities. Although informative, these approaches relied on static snapshots and neglected the temporal evolution of active regions.

Subsequent models introduced recurrent architectures to encode time dependency. Huang et al. (2018) employed LSTM-based methods to identify analogies between solar flare precursors and neuroscience time-series, while Kubo et al. (2020) demonstrated the use of CNNLSTM hybrids for time-evolving SHARP parameter sequences. However, LSTMs suffer from vanishing gradients and fixed memory scales, which limit their ability to capture long-range interactions especially when rare events exhibit subtle early precursors that unfold over several days.

87 Despite these advances, rare-event detection remains hindered by conventional architectures’ in-
 88 ability to handle extreme imbalance, to quantify uncertainty, and to prioritise temporally distributed
 89 precursor signals in a principled manner. These shortcomings motivate attention-based architectures
 90 with explicit mechanisms for signal compression, calibration, and tail modeling properties that are
 91 core to EVEREST.

92 **Transformers in Temporal Modeling** Transformer architectures’ use has expanded for time-
 93 series modeling due to their global attention mechanisms, which eliminate recurrence and enable
 94 flexible temporal context handling. Informer (Zhou et al., 2021) introduced a sparse self-attention
 95 mechanism to scale transformer models to long time series, while PatchTST (Nie et al., 2023) ex-
 96 tended this by modeling local patches across the time axis, improving forecast accuracy without
 97 increasing model depth. MEGA (Ma et al., 2023) further advanced temporal transformers by in-
 98 corporating moving-average features into the attention function to reduce volatility in sequential
 99 updates.

100 Despite their success, most transformer variants remain largely designed for regression tasks (fore-
 101 casting next values) or balanced classification, and often assume stationarity in the underlying dy-
 102 namics. Moreover, few integrate probabilistic outputs or attempt to explicitly reason about uncer-
 103 tainty. Crucially, standard transformers aggregate attention outputs either uniformly or via fixed
 104 positional priors, making them less effective at highlighting sparse, temporally localized precursors.

105 EVEREST addresses these limitations via a soft attention bottleneck—a learnable pooling layer that
 106 compresses the temporal latent space to a fixed embedding using relevance-weighted aggregation.
 107 This enables the model to extract predictive signatures from sparse precursors without requiring hard
 108 windowing or max-pooling, and is particularly well suited to early-event detection in non-stationary
 109 regimes.

110 **Uncertainty Quantification** Robust forecasting systems must not only predict whether an event
 111 will occur, but also express calibrated confidence in their outputs. In deep learning, predictive un-
 112 certainty is often decomposed into two parts: epistemic uncertainty, reflecting ignorance due to lim-
 113 ited data; and aleatoric uncertainty, reflecting inherent noise in observations. Classical approaches
 114 include Monte Carlo Dropout (Gal and Ghahramani, 2016), which approximates Bayesian infer-
 115 ence by averaging stochastic forward passes, and Deep Ensembles (Lakshminarayanan et al., 2017),
 116 which sample weights across multiple training runs.

117 These methods are computationally expensive and offer limited theoretical guarantees. To address
 118 this, evidential learning methods propose direct distributional parameterisation from data. Sensoy et
 119 al. (Sensoy et al., 2018), later extended by Van Amersfoort et al. (2021) and Charpentier et al. (2020),
 120 introduced methods for learning Normal-Inverse-Gamma (NIG) distributions, modeling the mean μ
 121 and variance σ^2 as random variables with higher-order uncertainty. The predictive distribution is:

$$p(y | x) = \int \mathcal{N}(y | \mu, \sigma^2) \cdot NIG(\mu, \sigma^2 | \theta) d\mu d\sigma^2$$

122 This framework enables simultaneous modeling of both aleatoric and epistemic uncertainty in a
 123 closed-form and is more efficient than sampling-based approaches. More recent work (Amini et al.,
 124 2020) has extended evidential learning to classification settings and demonstrated improved calibra-
 125 tion compared to temperature scaling and label smoothing.

126 EVEREST adopts the evidential regression head from Sensoy et al. (2018) to output four NIG param-
 127 eters per sample— (μ, v, α, β) —which define a full distribution over latent logits. The classification
 128 probability is obtained via $\sigma(\mu)$, i.e., a sigmoid applied to the predicted mean, allowing risk-sensitive
 129 learning while maintaining probabilistic interpretability.

130 Recent developments in distribution-free calibration, most notably conformal prediction, offer formal
 131 guarantees on predictive coverage without making strong assumptions about data distribution
 132 (Angelopoulos et al., 2021b; Romano et al., 2020). Although EVEREST does not yet incorpo-
 133 rate conformal methods directly, its evidential formulation shares conceptual goals with these ap-
 134 proaches, namely the production of calibrated probabilistic outputs under uncertainty. Integrating
 135 conformal prediction with evidential learning or EVT-based tail modeling represents a promising
 136 direction for future work, especially given the formal coverage guarantees shown by Angelopoulos
 137 et al. (2021a) and Romano et al. (2020), which connect conformal prediction with robust uncertainty
 138 estimation under weak assumptions.

Extreme Value Theory in Machine Learning Extreme Value Theory (EVT) provides a principled statistical framework for modeling the tails of distributions, particularly in high-stakes domains where rare but severe events matter. Classical results show that for sufficiently high thresholds u , the distribution of exceedances $y = x - u$ converges to the Generalized Pareto Distribution (GPD) (III, 1975). The GPD is defined by its shape ξ and scale σ :

$$P(y \leq t \mid y > u) = 1 - \left(1 + \xi \frac{t}{\sigma}\right)^{-1/\xi}, \quad \xi \neq 0$$

In practice, u is dynamically set as the 90th percentile of the logits within each training batch. This ensures adaptability to distributional shifts and avoids reliance on a fixed global threshold.

For comprehensive treatments of EVT in both theoretical and applied contexts, we refer readers to Coles (2001) and Embrechts et al. (1997), which serve as standard references in the field.

In machine learning, EVT has been applied to open-set recognition (Scheirer et al., 2014), anomaly detection (Zhao et al., 2020), and adversarial robustness (Cao et al., 2022). EVEREST extends this tradition by incorporating a lightweight tail-risk head that outputs the GPD parameters (ξ, σ) from the residual logits. During training, we apply a loss to penalize poor tail fits, improving the models ability to rank high-risk samples and maintain recall at low false-positive rates.

EVEREST Architecture The EVEREST model is designed as a lightweight yet expressive transformer-based sequence model tailored to rare-event time-series classification. It combines temporal compression, probabilistic uncertainty estimation, and tail-aware forecasting within a unified architecture.

Input Embedding and Positional Encoding Given an input sequence $x \in R^{B \times T \times F}$, where B is the batch size, $T = 10$ is the temporal horizon, and $F = 9$ denotes the number of SHARP-derived physical features (Abdullah et al., 2023), the input is first projected into a latent space of dimension $d = 128$ using a linear embedding layer:

$$h_0 = \text{LayerNorm}(\text{Linear}(x)) \in R^{B \times T \times d}.$$

To retain temporal structure, we apply sinusoidal positional encodings PE_t as in ?:

$$\hat{h}_0 = h_0 + \text{PE}, \quad \text{PE}_{t,2i} = \sin\left(\frac{t}{10000^{2i/d}}\right), \quad \text{PE}_{t,2i+1} = \cos\left(\frac{t}{10000^{2i/d}}\right).$$

Transformer Backbone The encoded sequence $\hat{h}_0$ is passed through $L = 6$ stacked transformer blocks, each consisting of multi-head self-attention (MHA), feedforward layers, and residual normalization:

$$h_{\ell+1} = \text{LayerNorm}(h_{\ell} + \text{Dropout}(\text{MHA}(h_{\ell})))$$

$$h_{\ell+1} = \text{LayerNorm}(h_{\ell+1} + \text{Dropout}(\text{FFN}(h_{\ell+1})))$$

where each $\text{FFN}(h) = \text{GELU}(\text{Linear}_1(h)) \rightarrow \text{Dropout} \rightarrow \text{Linear}_2$. This stack yields contextualized token representations $h_L \in R^{B \times T \times d}$.

Learned Attention Bottleneck To distill temporal information into a fixed-size summary, EVEREST introduces a soft attention bottleneck that learns a time-weighted aggregation over the sequence. Specifically, a single-head softmax attention weight vector $\alpha \in R^{B \times T \times 1}$ is computed as:

$$\alpha = \text{Softmax}(\text{Linear}(h_L)) \quad (\text{acrosstime}),$$

$$z = \sum_{t=1}^T \alpha_t \cdot h_{L,t} \in R^{B \times d},$$

where z serves as a temporally compressed representation summarising the entire input window. This soft aggregation enables the model to learn to attend to precursor patterns without requiring fixed thresholds or explicit alignment.

175 **Multiple Output Heads** From the latent summary z , EVEREST predicts a set of task-specific
 176 outputs via four separate heads:

177 **(1) Binary Classification Head.** A linear layer outputs a logit $l \in R$ for binary classification:

$$l = \text{Linear}_{\text{logits}}(\phi(z)), \quad \hat{y} = \sigma(l),$$

178 where ϕ denotes a dropout-regularized projection and $\sigma(\cdot)$ is the sigmoid function.

179 **(2) Evidential Head (NIG).** To quantify uncertainty, we parameterize a Normal-Inverse-Gamma
 180 (NIG) distribution over the latent logit:

$$(\mu, v, \alpha, \beta) = \text{Softplus}(\text{Linear}_{\text{nig}}(\phi(z))),$$

181 where μ is the mean, v the precision, and (α, β) control the higher-order uncertainty of the variance.

182 These parameters define a predictive distribution with closed-form variance:

$$\text{Var}(y) = \frac{\beta}{(\alpha - 1)v}, \quad \alpha > 1.$$

183 **(3) EVT Head (GPD).** To explicitly model tail risk, we add a head that outputs the shape ξ and scale
 184 σ parameters of a Generalized Pareto Distribution:

$$(\xi, \sigma) = \text{Softplus}(\text{Linear}_{\text{gpd}}(\phi(z))).$$

185 These define the tail distribution for residual exceedances beyond a fixed threshold u , using:

$$P(y > t \mid y > u) = \left(1 + \xi \frac{t - u}{\sigma}\right)^{-1/\xi}.$$

186 **(4) Precursor Head.** A lightweight scalar output indicates the presence of early-stage precursor
 187 patterns:

$$s = \sigma(\text{Linear}_{\text{precursor}}(\phi(z))) \in [0, 1],$$

188 allowing auxiliary training with early-time positive labels.

189 **Composite Loss Function** The total training loss is a convex combination of four components:
 190 a focal binary cross-entropy loss for classification, a negative log-likelihood (NLL) for the eviden-
 191 tial NIG head, a penalty for poor tail modeling via EVT, and a binary cross-entropy loss from the
 192 precursor detection head:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{focal}} + \lambda_2 \mathcal{L}_{\text{evidential}} + \lambda_3 \mathcal{L}_{\text{EVT}} + \lambda_4 \mathcal{L}_{\text{precursor}},$$

193 where $\lambda_1 = 0.8$, $\lambda_2 = \lambda_3 = 0.1$, and $\lambda_4 = 0.05$ in our experiments.

194 The evidential NLL is derived from the marginalization over the latent Gaussian, as shown in Sensoy
 195 et al. (2018). The EVT term penalizes misfit to the GPD distribution using exceedances over a
 196 threshold u .

197 **Training Stabilization and TSS-Based Early Stopping** To mitigate the impact of severe imbal-
 198 ance and unstable gradients, EVEREST incorporates three training stabilizers:

- 199 • **Gamma annealing.** The focal loss parameter γ is linearly annealed from 0 to a maximum
 200 $\gamma_{\text{max}} = 2.0$ over the first 50 epochs, helping to balance exploration and class separation.
- 201 • **Dynamic loss weighting.** The relative weights of the loss components are scheduled in
 202 three phases over training: an initial focus on the focal classification loss, followed by
 203 gradual introduction of evidential and EVT penalties. This stabilizes early learning before
 204 tail and uncertainty losses are introduced.
- 205 • **TSS-aware early stopping.** Model selection is based on the True Skill Statistic (TSS),
 206 defined as:

$$\text{TSS} = \frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}} - 1.$$

207 The best checkpoint is retained based on validation TSS, rather than loss or accuracy, to
 208 better reflect the rare-event detection objective.

209 This training regime enables EVEREST to maintain predictive sharpness while remaining robust
 210 under imbalance, class rarity, and non-stationarity.

211 3 Experiments

212 We can evaluate EVEREST on a solar flare forecasting benchmark constructed from SHARP phys-
 213 ical parameters and GOES flare labels (2010 - 2023). Our objectives are to assess EVEREST’s
 214 performance in rare-event detection, its robustness to class imbalance, its calibration under uncer-
 215 tainty, and its suitability for high-stakes operational forecasting.

216 **Datasets: SHARP-GOES** Our primary benchmark is constructed by aligning SHARP physical
 217 feature sequences with GOES X-ray flare reports (Abdullah et al., 2023). Each sample is a $T = 10$
 218 timestep window of $F = 9$ selected SHARP parameters (e.g., USFLUX, TOTUSJZ, TOTPOT)
 219 for a single NOAA active region, spaced at 12-minute intervals. To prevent potential data leakage
 220 across the training and testing phases, all samples are split on a per-region basis: active regions
 221 present in the test set are fully excluded from the training and validation sets. This ensures that
 222 model performance reflects generalization to unseen physical regions, not memorization of temporal
 223 fragments. Binary labels indicate whether a flare of a specified minimum class (C, M, or M5) occurs
 224 within a given horizon (24h, 48h, or 72h).

225 Training and test sets are stratified across all 9 (class and horizon) combinations. The resulting
 226 label distribution, highlights the severity of class imbalance, particularly for M5-class events (1:117,
 227 1:147, and 1:160 for 24h, 48h, and 72h horizons).

228 **Baselines** We compare EVEREST against published benchmarks drawn from recent peer-
 229 reviewed studies on solar flare forecasting using the SHARP-GOES dataset:

- 230 • **Liu et al. (2019):** Convolutional LSTM model with SHARP feature sequences for flare
 231 prediction (Liu et al., 2019).
- 232 • **Sun et al. (2022):** Time-series classifier with probabilistic balancing techniques for flare-
 233 level thresholds (Sun et al., 2022).
- 234 • **Abdullah et al. (2023):** Temporal transformer model trained on SHARP-GOES aligned
 235 sequences for multi-horizon forecasting (Abdullah et al., 2023).

236 Each method is evaluated under consistent data splits and horizon thresholds ($\geq C$, $\geq M$, and $\geq M5.0$
 237 classes at 24h, 48h, and 72h), as reported in the original studies. EVEREST is benchmarked under
 238 consistent settings to ensure comparability, with results summarised in Figure 1.

239 **Evaluation Metrics** We adopt a range of metrics to evaluate both classification and calibration
 240 performance under rare-event conditions:

- 241 • **True Skill Statistic (TSS):** difference between hit-rate and false-alarm rate; remains infor-
 242 mative under extreme class imbalance.
- 243 • **Precision (positive predictive value):** fraction of predicted positives that are correct.
- 244 • **Recall (sensitivity):** fraction of actual positives that are detected.
- 245 • **F1 Score:** harmonic mean of Precision and Recall, balancing over- and under-prediction.
- 246 • **Brier Score** (the lower the better): mean-squared error of probabilistic predictions.
- 247 • **Log-loss** (the lower the better): negative log-likelihood of the predicted probabilities, pe-
 248 nalising over-confident errors.

249 **Results: Solar Flare Forecasting** EVEREST achieves high TSS across all horizons and flare
 250 intensities, particularly for rare M5-class events as can be seen in Figure 1. Its calibrated uncertainty
 251 outputs and attention bottleneck contribute to stable recall despite extreme imbalance.

252 **Ablation Studies** Table 1 summarizes ablations for M5-class, 72-hour forecasting. The *No EVT*
 253 *head* variant achieves the highest TSS (0.960) and recall (0.962) but with the lowest precision
 254 (0.532), indicating overprediction. *Focal-only loss* offers the best recall (0.971) and competitive
 255 TSS (0.951), balancing sensitivity and sharpness. Removing the precursor head yields the highest
 256 calibration (lowest Brier 0.00072 and log-loss 0.00656) with strong F1 (0.821) and TSS (0.923).
 257 *Mean pooling* performs similarly in TSS (0.922) but with lower precision. Removing the evidential

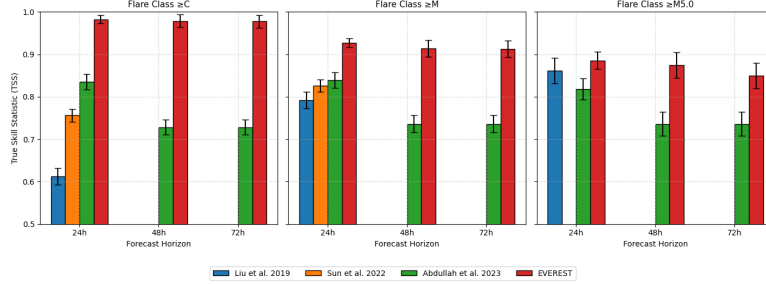


Figure 1: TSS Comparison Across Models and Flare Classes

Table 1: Ablation study on M5-class, 72 h horizon. Bold indicates the best (highest, lowest) value in each column.

Variant	Prec.	Rec.	F1	Brier	Log-loss	TSS
Full model	0.79167	0.91346	0.84821	0.00107	0.02278	0.91311
No EVT head	0.53191	0.96154	0.68493	0.00114	0.01004	0.96031
No evidential head	0.74336	0.80769	0.77419	0.00102	0.00902	0.80729
No precursor head	0.73846	0.92308	0.82051	0.00072	0.00656	0.92260
Mean pooling	0.65753	0.92308	0.76800	0.00117	0.01134	0.92238
Focal-only loss	0.72662	0.97115	0.83128	0.00086	0.00875	0.95062
No gamma annealing	0.66400	0.79808	0.72489	0.00066	0.00315	0.79749

head shows moderate degradation (TSS 0.807), while *no gamma annealing* leads to the weakest overall performance. The full model delivers the strongest F1 (0.848) and highest precision (0.792), maintaining robust calibration and balanced skill across metrics.

4 Discussion

Interpretable decisionmaking and calibrated uncertainty estimation are critical in highrisk forecasting domains such as space weather and autonomous system diagnostics. In EVEREST, interpretability arises from both the temporal attention mechanism and the structured probabilistic outputs from the evidential and EVT heads. This section explores these components qualitatively and quantitatively.

Attention Heatmaps The attention bottleneck in EVEREST enables temporal interpretability by assigning scalar weights to each timestep in the input sequence. These learned weights highlight regions of precursor activity that most contribute to the models final decision. Since the attention pooling is implemented via a softmax layer over linear projections, the weights are bounded, differentiable, and fully learnable.

To examine attention behaviour, we visualise the attention maps α_t across the $T = 10$ timesteps for selected samples. Figure 2 shows four representative cases arranged in a 2×2 grid.

As seen in Fig. 2, true positives (panel a) exhibit sharply peaked midwindow attention, true negatives (panel b) distribute attention more evenly, false positives (panel c) focus on spurious midwindow fluctuations, and false negatives (panel d) underattend critical latestage signals. These patterns illustrate how EVERESTs attention bottleneck captures meaningful temporal cues aligned with prediction outcomes.

Evidential Uncertainty EVERESTs evidential head outputs four NIG parameters (μ, v, α, β) per sample, from which we compute total, epistemic and aleatoric variance:

$$\text{Var}_{\text{total}} = \frac{\beta}{(\alpha - 1)v}, \quad \text{Var}_{\text{epistemic}} = \frac{\beta}{(\alpha - 1)v\alpha}, \quad \text{Var}_{\text{aleatoric}} = \frac{\beta}{(\alpha - 1)v} (1 - 1\alpha).$$

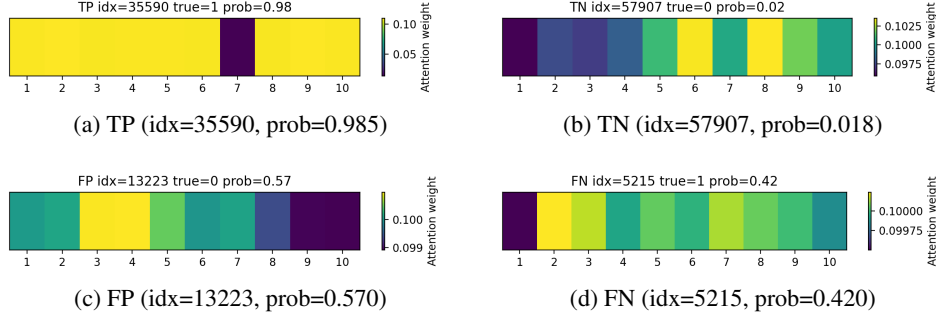


Figure 2: attentionweight heatmaps over 10 timesteps for Cclass, 72h flare forecasting. (a) true positive, (b) true negative, (c) false positive, (d) false negative. Colours denote the softmaxnormalized attention weight at each timestep (110). Samples were chosen to have predicted probabilities closest to the median of their outcome group.

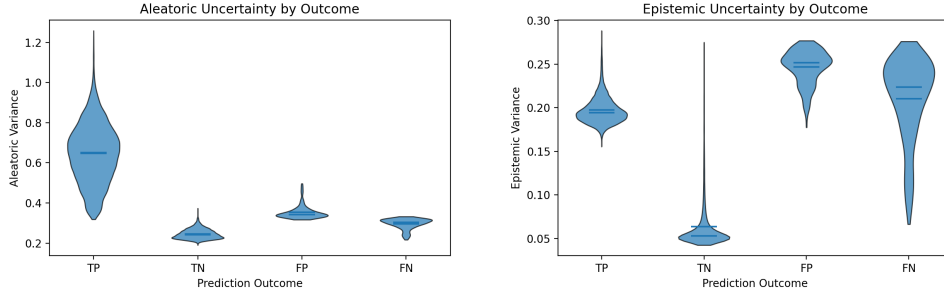


Figure 3: (a) aleatoric variance by outcome and (b) epistemic variance by outcome, for true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). Medians are indicated by the thick lines inside each violin.

281 To assess whether uncertainty correlates with prediction errors, we group test samples by outcome
 282 (TP, TN, FP, FN) and plot the resulting variance distributions in Fig. 3.

283 From panel (a), aleatoric variance is highest for TP samples (median 0.65), reflecting true mea-
 284 surement noise in flaring regions, and lowest for TN (median 0.23). Panel (b) shows epistemic
 285 variance peaks on misclassified examples (FP median 0.25, FN median 0.22) and is near zero for
 286 TN (median 0.05), confirming that EVEREST’s evidential head signals low confidence when it errs.

287 **EVT Head Tail Sensitivity** EVEREST’s EVT head parameterizes the distribution of exceedances
 288 via a Generalized Pareto Distribution (GPD), outputting a shape ξ and scale σ per sample. To
 289 summarize tail risk, we define the *EVT score* as

$$s_{\text{EVT}} = \xi \cdot \sigma,$$

290 where higher values indicate heavier or more volatile tails and thus greater inferred probability of
 291 extreme flares.

292 Figure 4 shows s_{EVT} distributions across GOES flare classes (C, M, M5). We observe a clear
 293 upward trend in medians and interquartile ranges, with Spearman $\rho = 0.752$ and Kendall $\tau = 0.613$,
 294 confirming that EVT scores correlate with flare severity.

295 EVEREST demonstrates that principled architectural design grounded in probabilistic reasoning and
 296 calibrated attention mechanisms can substantially improve rare-event sequence classification under
 297 extreme class imbalance and distributional uncertainty.

298 In addition to performance, EVEREST offers auxiliary interpretability via its attention bottleneck,
 299 which assigns relevance scores to temporally localized features. While not inherently explainable,

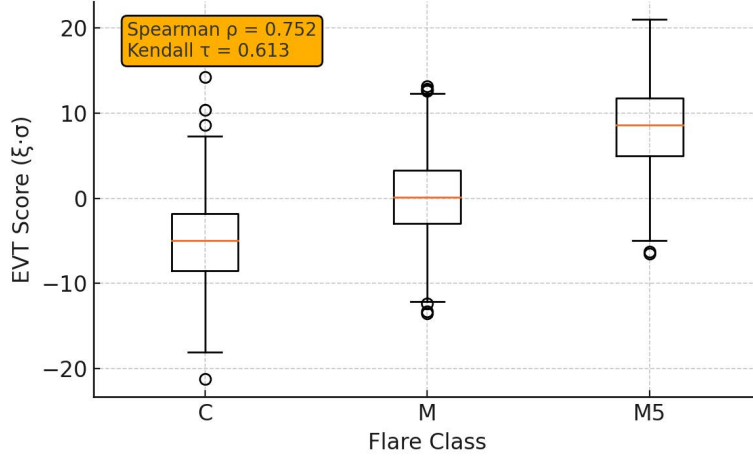


Figure 4: EVT score distributions ($\xi \cdot \sigma$) by flare class (C, M, M5). Medians rise from left to right, indicating heavier inferred tails for more extreme flares. Spearman $\rho = 0.752$, Kendall $\tau = 0.613$.

300 this mechanism provides actionable insight into the models decision process-particularly valuable
 301 for domain experts analyzing precursor dynamics in time-series anomalies.

302 The evidential head outputs structured uncertainty estimates, capturing both epistemic and aleatoric
 303 components in a fully differentiable manner. This contrasts with traditional confidence scores or
 304 sampling-based approximations, which often fail under imbalance. In parallel, the EVT head explic-
 305 itly models tail risk via a learned Generalized Pareto distribution, enhancing recall on high-severity,
 306 low-frequency events such as M5-class flares.

307 EVEREST is also efficient and compact, with fewer than 3 million parameters and the ability to train
 308 on commodity hardware (e.g., Apple M2 MacBook Pro) without GPU acceleration. This makes it
 309 practical for near-edge deployment in domains like satellite telemetry, grid monitoring, and biomed-
 310 ical diagnostics.

311 Although developed for solar flare forecasting, EVERESTs inductive biases-temporal compression,
 312 uncertainty estimation, and tail modeling-are broadly applicable. Its modular architecture enables
 313 transfer to other high-stakes settings without reliance on domain-specific tuning.

314 5 Conclusion and Limitations

315 We presented **EVEREST**, a transformer-based architecture for calibrated, interpretable, and tail-
 316 aware rare-event forecasting. The model combines a learnable attention bottleneck, a Normal-
 317 Inverse-Gamma head for evidential uncertainty, a Generalized Pareto head for tail modeling, and
 318 a precursor scoring module for early detection-all trained jointly via a composite loss that balances
 319 classification, calibration, and tail risk.

320 EVEREST outperforms prior models on a decade-long solar flare forecasting benchmark, achieving
 321 state-of-the-art TSS across 24, 48, and 72-hour horizons.

322 Despite its strong performance, EVEREST has limitations. It relies on fixed-length inputs and as-
 323 sumes temporally aligned, i.i.d. samples, restricting its applicability to streaming or asynchronous
 324 settings. It also operates solely on tabular features, omitting spatial modalities like magnetograms.
 325 Inference latency for longer horizons remains unprofiled.

326 Future work will extend EVEREST to support streaming inference, multimodal inputs, and active
 327 learning based on uncertainty and tail risk. By bridging attention-based modeling with principled
 328 statistical inference, EVEREST offers a robust foundation for high-stakes forecasting in scientific
 329 and industrial domains.

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390 NeurIPS Paper Checklist

391 1. Claims

392 Question: Do the main claims made in the abstract and introduction accurately reflect the
393 paper’s contributions and scope?

394 Answer: [Yes]

395 Justification: The abstract and introduction accurately summarize the papers architectural
396 contributions, including the attention bottleneck for latent temporal compression, the evi-
397 dential head for calibrated uncertainty estimation via NIG, and the EVT head for explicit
398 tail risk modeling. These components are jointly optimized and evaluated in the experi-
399 mental sections, with performance supported by TSS and calibration metrics (Tables ??,
400 1). Section 4 further substantiates the models interpretability and probabilistic expressive-
401 ness through attention heatmaps and variance decomposition. While results are reported
402 on solar flare data, the abstract, introduction, and conclusion clearly state that EVEREST’s
403 contributions lie in the modeling paradigm, not in domain specialization. The architecture
404 is presented as a modular template for rare-event detection under distributional shift and
405 imbalance, and this scope is consistently maintained throughout the paper.

406 2. Limitations

407 Question: Does the paper discuss the limitations of the work performed by the authors?

408 Answer: [Yes]

409 Justification: Section 6 discusses limitations of the EVEREST architecture, including its
410 reliance on fixed-length windows and its current lack of support for variable-length or
411 streaming sequences both of which affect generalizability to asynchronous or real-time de-
412 ployments. The models use of numerical input features excludes spatial modalities, which
413 may limit performance in domains where spatial structure is key. Further, while the ar-
414 chitecture is modular, it is only evaluated on a single scientific dataset, so cross-domain
415 generalization remains unverified. Training efficiency on CPU-class hardware is demon-
416 strated, but inference costs for longer horizons are not profiled. Assumptions such as i.i.d.
417 window sampling and aligned temporal context may not hold in more complex or noisy
418 environments. While not a fairness- or privacy-critical model due to the physical nature of
419 the data, broader deployment contexts may require addressing those concerns. These limita-
420 tions are architectural and methodological, and they motivate extensions toward streaming,
421 multimodal, and cross-domain robustness.

422 3. Theory assumptions and proofs

423 Question: For each theoretical result, does the paper provide the full set of assumptions and
424 a complete (and correct) proof?

425 Answer: [NA]

426 Justification: This paper does not include formal theoretical results, theorems, or proofs.
427 The contributions are architectural and empirical in nature, focusing on a modular
428 transformer-based framework for rare-event sequence modeling. While the loss formu-
429 lations (e.g., evidential NLL, EVT penalty) and distributional choices (NIG, GPD) are
430 grounded in established probabilistic theory, the paper does not introduce new theorems
431 or attempt novel theoretical derivations. All equations provided are implementation-level
432 expressions to clarify model behavior and optimization not propositions requiring formal
433 proof. As such, no theoretical assumptions or proofs are applicable.

434 4. Experimental result reproducibility

435 Question: Does the paper fully disclose all the information needed to reproduce the main
436 experimental results of the paper to the extent that it affects the main claims and/or conclu-
437 sions of the paper (regardless of whether the code and data are provided or not)?

438 Answer: [Yes]

439 Justification: The paper provides all architectural, training, and evaluation details needed
440 to reproduce the main results. Section 3 outlines the full model design, including all sub-
441 modules, activation functions, and loss components. Section 4 describes the dataset con-
442 struction pipeline from SHARP and GOES sources, with explicit forecasting horizons, flare

thresholds, and splits per class. Section 4 reports evaluation metrics and ablation protocols. Code is included in the submission too. The paper includes sufficient implementation-level details for independent reproduction, including the focal loss variant, NIG and GPD parameterizations, and training stabilizers such as gamma annealing and TSS-based early stopping. Any further details - e.g., precise optimizer configuration or random seed control - these will be included in the final supplemental material (released codebase).

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [\[Yes\]](#)

Justification: The SHARPGOES dataset used in this study is publicly available and originally prepared in Abdullah et al. (2023), with links and access instructions included in the supplemental material. We use this exact data source with consistent preprocessing and time alignment protocols. The full implementation of EVEREST including architecture, loss functions, and evaluation routines will be released in the codebase, with a reproducibility-friendly folder structure and a configuration-driven pipeline. The final supplemental material will include training scripts, environment specifications, and README documentation outlining how to reproduce the results.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [\[Yes\]](#)

Justification: Section 3 describes the model architecture and training setup, including attention bottlenecks, loss function composition, and evaluation metrics. Appendix Table 1 lists all hyperparameters such as embedding dimension, number of attention heads, optimizer configuration, gamma annealing schedule, and focal/evidential/EVT loss weights. The training and test splits exactly follow those used by Abdullah et al. (Abdullah et al., 2023), as discussed in Section 3. Although the detailed search strategy is not reported in the main text, the provided codebase includes full hyperparameter settings. Thus, sufficient detail is given to understand and reproduce the results.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [\[Yes\]](#)

Justification: We report standard deviation error bars for the EVEREST models TSS across three random seeds per configuration in Figure 1, with the source of variability being random weight initialization and training data shuffling. Error bars are denoted in parentheses in the Appendix, e.g., **0.9799** (0.01), and reflect the standard deviation of final TSS across seeds. While we do not assume normality of errors, we follow the community norm of reporting 1-sigma deviations. Additional robustness is assessed through ablation studies (Table 1), which qualitatively confirm the stability of learned distributions. All metrics are computed on fixed train/test splits provided by Abdullah et al. (2023), ensuring consistency across runs. Error estimation logic is implemented via NumPy and PyTorch using deterministic seeds, with full reproducibility described in the training code and Appendix.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [\[Yes\]](#)

Justification: The experiments were conducted on two compute environments: an NVIDIA Ada A6000 GPU server and an Apple M2 MacBook Pro using MPS acceleration. Each EVEREST model (per class and forecasting horizon) required up to 1 hour of training on

the A6000, depending on sample count and class imbalance severity. On the MacBook Pro, training times ranged between 23 hours per configuration due to reduced throughput. All experiments used mini-batches of 512, mixed-precision training (AMP), and consistent data splits. Only the reported runs were used for final evaluation; no additional exploratory sweeps or discarded training runs are omitted from reporting. These compute details are documented in Appendix and are reproducible using the provided code and instructions.

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The research complies with the NeurIPS Code of Ethics in all respects. It does not involve human subjects, sensitive personal data, or dual-use technologies with foreseeable potential for misuse. The data used (SHARP parameters and GOES flare labels) are publicly available from NOAA and NASA repositories, and the analysis focuses solely on forecasting solar flares as an astrophysical phenomenon. All code development and evaluation were conducted with transparency, fairness, and environmental impact in mind, including resource-efficient training on commodity hardware. Anonymity and responsible conduct have been preserved throughout the submission process.

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The paper discusses both potential benefits and limitations of deploying the EVEREST architecture in high-stakes forecasting scenarios. On the positive side, EVEREST is designed to improve early warning and uncertainty-aware decision-making in domains such as space weather, aviation, and satellite operations, potentially reducing disruptions and economic loss from rare but severe events. The model's design prioritizes interpretability and calibrated confidence, supporting human oversight. On the risk side, the paper acknowledges that high-confidence forecasts might be misused if acted upon without human verification or contextual understanding. EVEREST was evaluated solely on solar flare forecasting and does not currently address fairness, privacy, or misuse risks in broader applications. However, its architecture is domain-agnostic and could be deployed in safety-critical areas; future work should explore governance mechanisms, such as usage audits or uncertainty thresholds, to prevent overreliance. Overall, the paper reflects on the societal implications of predictive modeling under uncertainty, balancing utility with caution.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [NA]

Justification: The EVEREST model is designed for scientific forecasting of solar flares using publicly available NASA/NOAA data and poses no foreseeable risk of misuse or dual-use harm. The data are physical in nature (solar magnetic field parameters) and not subject to privacy, surveillance, or generation risks. As such, the paper does not involve the release of high-risk models or datasets that would necessitate additional safeguards.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The primary dataset used in this study is SolarFlareNet, published by Abdulla et al. (2023) and hosted at <https://nature.njit.edu/solardb/solarflarenet>. The data is derived from NASA's SDO/HMI instrument, which is in the public domain. While

the SolarFlareNet website does not explicitly state a license, the dataset is publicly accessible and intended for scientific research. We properly cite the datasets creators in the main text and references, adhering to the attribution guidelines implied in the associated Scientific Reports article. All other components of the work, including model code and training procedures, were developed by the authors.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Answer: [Yes]

Justification: This work uses the SolarFlareNet dataset introduced by ?, available via the NJIT-hosted portal at <https://nature.njit.edu/solardb/solarflarennet>. While no license is explicitly stated, we use the dataset strictly for academic, non-commercial purposes and properly attribute the source. If required, we are prepared to contact the maintainers for further licensing clarification. No additional datasets or assets are released as part of this work.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: This paper does not involve any form of crowdsourcing or research with human subjects. All data used are publicly available from scientific repositories and do not require interaction with human participants.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: The research does not involve human subjects and therefore did not require IRB or equivalent ethical review. All experiments were performed on publicly available physical datasets.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [NA]

Justification: Large language models were not used for any core component of method development, experimentation, or data processing. Any LLM use was limited to standard writing support and not relevant to the scientific contribution.